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Graduation Project Report

Submitted for the obtention of:

The National Engineering Diploma
In Applied Sciences and Technology

Title:

Mechanical and Automation Engineering of Stacker Crane for Crimp Applicators

Hosting Company:

DRÄXLMAIER GROUP SOUSSE

Presented on the: 06/30/2018

Author: *Amani GUEFRECH*

INSAT, Automation Control and Industrial Computing

Professional Tutor: *Younes RGUEZ*
Maintenance Department Manager

Academic Tutor: *Raouf KETATA*
University Lecturer

Co-Tutor*: Atef BOULILA*
Assistant professor

Academic Year: 2017-2018

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DEPOSITION AUTHORIZATION

Company Supervisor	Academic Supervisors
Date: …/…/2018	Date: …/…/2018
(Seal and Signature are mandatory)	(Signatures are mandatory)

Academic Year: 2017-2018

Dedication

It is my genuine, gratefulness and warmest regard that I dedicate this dissertation to my beloved people who contributed in a way or another to my success:

My father's memory *who would have been happy to see me today, may you find peace and happiness in paradise. I hope I will live up for your expectations.*

Mum*, whose unconditional love knew no bounds. No words could express my pride and gratitude toward your sacrifices You was my source of strength and inspiration who taught me to work hard for the things that I aspire to achieve. I want to thank you for all you have done and wish you a good health, prosperity and happiness*

Mayssa*, my little sister, thank you for being there for me with your sincere love and endless support. I wish you the best luck for your bright and prosperous future.*

My grandparents, ***Mustafa, Jamila*** *and* ***Aycha****, I owe so much to you for your undying support and advice. I thank you for believing in me and for giving motivations to go ahead.*

All my family*, I cannot express my gratitude and pride for yours love and pride.*

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Contents

FIGURES LIST

TABLES LIST

ABBREVIATIONS

GENERAL INTRODUCTION.....1
CHAPTER 1: GENERAL CONTEXT AND PROBLEMATIC STATEMENT.....3
INTRODUCTION 4
1. HOSTING COMPANY 4
1.1. Dräxlmaier Group 4
1.2. Sales and Clientele.....5
1.3. METS.....5
2. PROJECT PRESENTATION 10
2.1. Project General Context.....10
2.2. Project Requirements Specification.....10
2.3. Adopted Approach.....11
2.4. Project Charter 11
CONCLUSION 13
CHAPTER 2: CURRENT STATE ANALYSIS AND PROPOSED SOLUTION.....14
INTRODUCTION 15
1. CURRENT STORAGE METHODOLOGY 15
1.1. Tools Warehouse 15
1.2. Crimp Applicator's Specifications 15
1.3. Storage System.....17
2. PROPOSED SOLUTION 23
2.1. Automated Stacker Crane 23
2.2. Horned Beast Flowchart 23
2.3. Octopus Diagram 24
2.4. FAST Analysis.....26
CHAPTER3: MECHANICAL AND ELECTRICAL DESIGN 29
INTRODUCTION 30
1. MECHANICAL DIMENSIONING 30
1.1. Technological Concept Diagram 30

1.2. Storage Area.....30
1.3. Retrieval Function (Y).....31
1.4. Lifting Function (Z).....32
1.5. Gripping Function (X).....33
2. INSTRUMENTATION.....36
2.1. Actuators dimensioning.....36
2.2. Pre-actuators specifications.....38
2.3. Switching Sensors.....39
3. PROCESS AND CONTROL SPECIFICATIONS.....40
3.1. PLC characteristics.....40
3.2. HMI characteristics.....41
4. ELECTRICAL INSTALLATION.....42
CONCLUSION.....42
CHAPTER 4: STACKER CRANE AUTOMATION AND HMI DEVELOPMENT.....42
INTRODUCTION.....44
1. AUTOMATION.....44
1.1. General description.....44
1.2. Hardware Configuration.....47
1.3. Network configuration.....48
2. HMI CONCEPTION.....49
2.1. Stacker Crane Use Case Diagram.....49
3. IHM DEVELOPMENT.....51
3.1. Registration.....51
3.2. Insert New Crimp Applicator.....52
3.3. Rack_A or Rack_B.....52
3.4. Control_System.....53
CONCLUSION.....53
GENERAL CONCLUSION.....55
NETOGRAPHY.....56
BIBLIOGRAPHY.....56
APPENDIX

Figures List

Figure 1: Geographical Distribution of Dräxlmaier Subsidiaries 4

Figure 2: DRX Customers 5

Figure 3: Sales Evolution 5

Figure 4: Komax Alpha 488 Crimping Machine-Verdrillen / Wire Twister 6

Figure 5: Design 1/1 for BM4B407/117 Motor Module 7

Figure 6: Order Flow: Customer Supplier 8

Figure 7: METS Data Flow Diagram.....9

Figure 8: METS Materials Flow Diagram 9

Figure 9: Tools Warehouse CatiaV5 3D Drawing.....10

Figure 10: RACI Categories 13

Figure 11: Managed Tools 15

Figure 12: Referenced Racks 17

Figure 13: Safety and Security Equipment's 18

Figure 14: Illegible Reference 19

Figure 15: Hand-Written Reference.....19

Figure 16: Non-Respect of Storage Decrescent Order 19

Figure 17: Tools defected Label 20

Figure 18: Repair and Maintenance Area 20

Figure 19: Ichikawa Diagram 21

Figure 20: RULA score for 1st Critical Posture 22

Figure 21: RULA score for 2nd Critical Posture 22

Figure 22: Horned Beast Diagram 23

Figure 23: Octopus Diagram.....24

Figure 24: Pareto Diagram 25

Figure 25: FAST Diagram 28

Figure 26: Stacker Crane Technological Concept 30

Figure 27: HepcoMotion TS -350 proposed dimensions 31

Figure 28: HDS2 Catia 3D Design 32

Figure 29: Hepcomotion HB proposed specification.....32

Figure 30: Catia V5 Inertia Measure 33

Figure 31: Rack Driven Carriage 3D design.....33
Figure 32: Terminal Strip Positionning.....34
Figure 33:DSNU -20-250-P-A.....36
Figure 34:EF300CTC electromagnet lock.....37
Figure 35: 5/2 distributor.....38
Figure 36: Choosing sensors Methodology.....39
Figure 37: Visual C#.....42
Figure 38: Management of different Modes.....44
Figure 39: Security "GRAFCET".....45
Figure 40: Management of Storage and retrieval GRAFCET.....45
Figure 41: Operating tools retrieval and storage.....46
Figure 42: Developped macro-steps.....47
Figure 43: Hardware configuration.....48
Figure 44: Network configuration.....48
Figure 45: Use Case Diagram.....49
Figure 46: Log in Interface.....51
Figure 47: Insert_new Interface.....52
Figure 48: Rack_A Interface.....52
Figure 49: Control interface.....54

Tables List

Table 1: METS Fact Sheet 5
Table 2: Project Charter 12
Table 3: Comparison Between KS and VK Status 16
Table 4: Detailed Tools Characteristics 16
Table 5: Risk Assessment at Tools' Warehouse 18
Table 6: RULA Scoring 21
Table 7: Paired Comparison Matrix 25
Table 8: Comparison between PLC and Microcontroller 26
Table 9: Comparison between movement Transformation mechanism 27
Table 10: Initial Parameters 34
Table 11: Positioning information 37
Table 12: Motors dimensioning 38
Table 13: S7 PLC's Comparison 40
Table 14: INPUT/OUTPUT classification 41
Table 15: Recommended Input and Output modules 47

Abbreviations

BRT	: Bereit **(Gathering area)**
DMAIC	: Define, Measure, Analyze, Innovate/Improve, Control
DRX	: Dräxlmaier
EE	: Electrical Energy
FAST	: Function Analysis System Technique
FBB	: FormBrettBau (**Board Construction**)
FST	: Fertigung Steurung **(Production Control)**
GRAFCET	: Functional Graph of Control Step Transition
HMI	: Human-Machine Interface
JIS	: Just-In-Sequence
JIT	: Just-In-Time
KS	: Kabel Schneiderei (Electrical Wire Cut)
KSK	: Kundenspezifischer Kabel baum (**Custom Cable Tree)**
ME	: Mechanical Energy
METS	: Electrotechnical Manufacture of Sousse
PLC	: Programmable Logic Controller
RULA	: Rapid Upper Limb Assessment
TPM	: Technische ProduktionsManagement **(Production Management Techniques)**
VK	: Vor Konfektion **(Pre-Confection)**

General Introduction

Machinery and warehousing technology have had significant improvement since the seventies of last century. In addition to manually powered material handling solutions, mechanical and highly automated handling systems have also appeared. Warehouses with automated storage/retrieval system have rapidly spread due to the capability of storing a large amount of goods or tools in a small area.

Not only automotive industries are continually improving and taking advantage of these technological advances to grab market shares but also their suppliers. Driven by globalization and increasing customer requirements, they are forced to offer a large range of vehicle models and options. They are strengthening their production procedures to meet their customer needs: product conformity to certain standards of quality and respecting Lead time.

Electro technical Manufacture of Sousse METS, one of automotive suppliers specialized in wiring harness manufacturing, opts for optimizing its production management to guarantee flexibility. Consequently, different departments are taking parts in the manufacture vision especially Warehouses whose role is necessary to assume the success of Just-In-Time (JIT) production.

Warehouses can be dedicated to raw material, tools, spare parts or finished products storage. At METS, Tools-Warehouse is the crimp applicator's supplier to Electrical Wire Cut (KS) and Pre-Confection (VK) production.

This graduation project aims to improve the storage management of these tools. It consists in designing and automating a Stacker Crane for Crimp Applicators to facilitate its handling and reorganize its flow.
The graduation project report is divided into 4 chapters:

- The first chapter « General context and Problem statement » gives a brief overview of METS company and explains the problem treated throughout the project.

- The second one « Current state analysis and proposed solution » is devoted to explaining the current storage procedure at tools-warehouse and its deficiencies as well as well introducing the proposed solution.

- The third chapter « Mechanical and Electrical design» is dedicated to detail the mechanical and electrical chosen solutions and to propose the electrical wiring diagrams.

- The fourth and last one « Stacker Crane Automation and HMI Development » presents automation specifications and the control through Human-Machine Interface.

Chapter 1: General Context and Problematic Statement

Introduction

In this chapter, a general context of the graduation project is presented. I start by introducing the hosting company “METS”, the motivations leading to this project and finish by explaining the adopted approach and detailing the work schedule.

1. Hosting Company

1.1. Dräxlmaier Group

Dräxlmaier Group is a German industrial group founded in 1958 in Vilsbiburg by Herr Fritz Dräxlmaier. It is a supplier for the premium automotive industry. It has more than 60 production sites present in over 20 countries throughout the world covering the 5 continents (**Figure 1**)



Figure 1: Geographical Distribution of Dräxlmaier Subsidiaries [1]

Its main business activities concentrate on designing, developing, JIT and JIS testing as well as manufacturing high-end automotive interior equipment:

- Electrical components: Wiring harnesses, Modular Power Distributors, Connector systems, Contacts,
- Electronic components: Battery management systems, Intelligent power distributors, Interior and Body electronics,
- Interiors: Door Panels, Ambient lighting, Consoles, Instrument panels,
- Battery systems: Low-Voltage Battery Systems, High-Voltage Battery Systems, High-Voltage Control boxes.

1.2. Sales and Clientele

DRX Group counts around 70,000 associates around the world. It is ranked among the top 100 global automotive suppliers. Group sales grew by 29% in the last 5 years and reached 4100 million EUR in 2017; the representations below are about the group's clientele (**Figure 2**) and volume of sales in million EUR (**Figure 3**).



Figure 2: DRX Customers

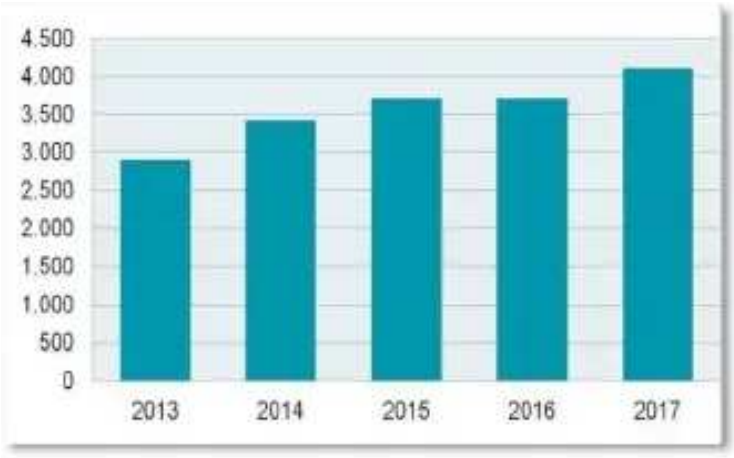


Figure 3: Sales Evolution [1]

1.3. METS

Electrotechnical Manufacture of Sousse is a German limited partnership company. It's among the four subsidiaries of DRX in Tunisia. The factory is a total exporter specialized in producing high-end cars wiring harnesses.

1.3.1. Fact Sheet Of METS

The presentation of METS plant is detailed in the sheet below (**Table 1**):

Table 1: METS Fact Sheet

Dräxlmaier Group Sousse (METS)	DRÄXLMAIER
Holding Company: DRÄXLMAIER Group Address: Industrial zone of Sidi Abdelhamid 4061 Sousse Sector: Automotive subcontracting Products: Wiring harnesses Telephone: (+216) 31 26 04 01	Fax: (+216) 73 32 22 34 Date of Creation: 1974 Staff (in 2018): 2500 employee Certifications: ISO14001, IATF16949, VDA6.3 Plant Manager: Anis KAMMOUN Website: www.draexlmaier.tn/

1.3.2. Product Range

DRX Sousse manufacture respects customer technical specifications. It assures the production of engines and panels modules for automotive industries:

- Production of modules for assembly in Germany,
- Production of modules for assembly in Tunisia,
- Production of modules and direct supply to customer.

1.3.3. Production Departments

DRX Sousse production combines four principal departments: KS, VK, Muster Bau and Assembly and KSK.

a. KS (Electrical Wire Cut)

Four phases are carried out during the first stage of manufacturing (**Figure 4**):

- Cutting Wire: according to a specific dimension,
- Stripping: it can be a unique or double removing insulation,
- Crimping: wire and electrical connector assembly,
- Twisting of the two Wire.



Figure 4: Komax Alpha 488 Crimping Machine-Verdrillen / Wire Twister [2]

b. VK (Pre-Confection)

It's a cable preparatory activity for final stage. It consists in performing one of the following tasks with semi-automatic machines:

- "U-Schall": for ultrasonic cable welding,
- "Verdrillen" (twist in English): for 2 or 3 cable twisting,
- "Schlauch Schneiden" (crosscutting flexible tube in English): for cutting pipes,
- "Crimpen" for cable cutting, stripping, crimping.

c. Muster Bau (Prototype Manufacturing)

It is an overriding stage including:

- Special product manufacture (prototypes, pre-production parts…),
- Spare parts manufacture (these are products already manufactured on the chain and ordered by the customer as spare parts in small quantities),
- Test of new products.

d. Assembly and KSK (Custom Cable Tree) departments

It assures cables and accessories assembly according to design 1/1 already fixed on FBB (Board Construction) work boards (**Figure 5**) to supply finished-products delivered to customers or semi-finished products for assembly in other subsidiaries. Assembly areas are defined by the type of product (PQ46, PQ47, AB2 ...) and are equipped with chains for assembling large modules or fixed positions dedicated to relatively small or medium modules.

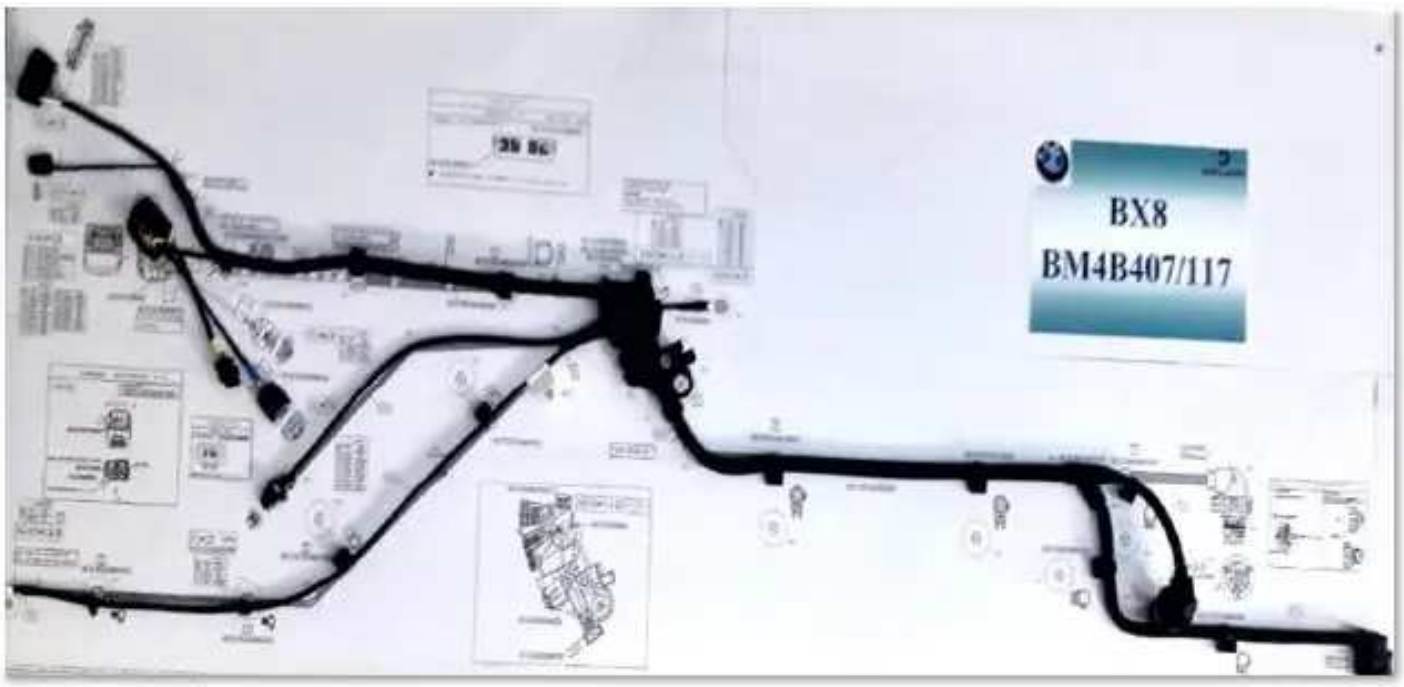


Figure 5: Design 1/1 for BM4B407/117 Motor Module

1.3.4. METS Flow

a. Customer Order Flow

METS customer orders are managed through holding company DRX Group. The process trigger is generated by the sales order as presented in **Figure 6**.

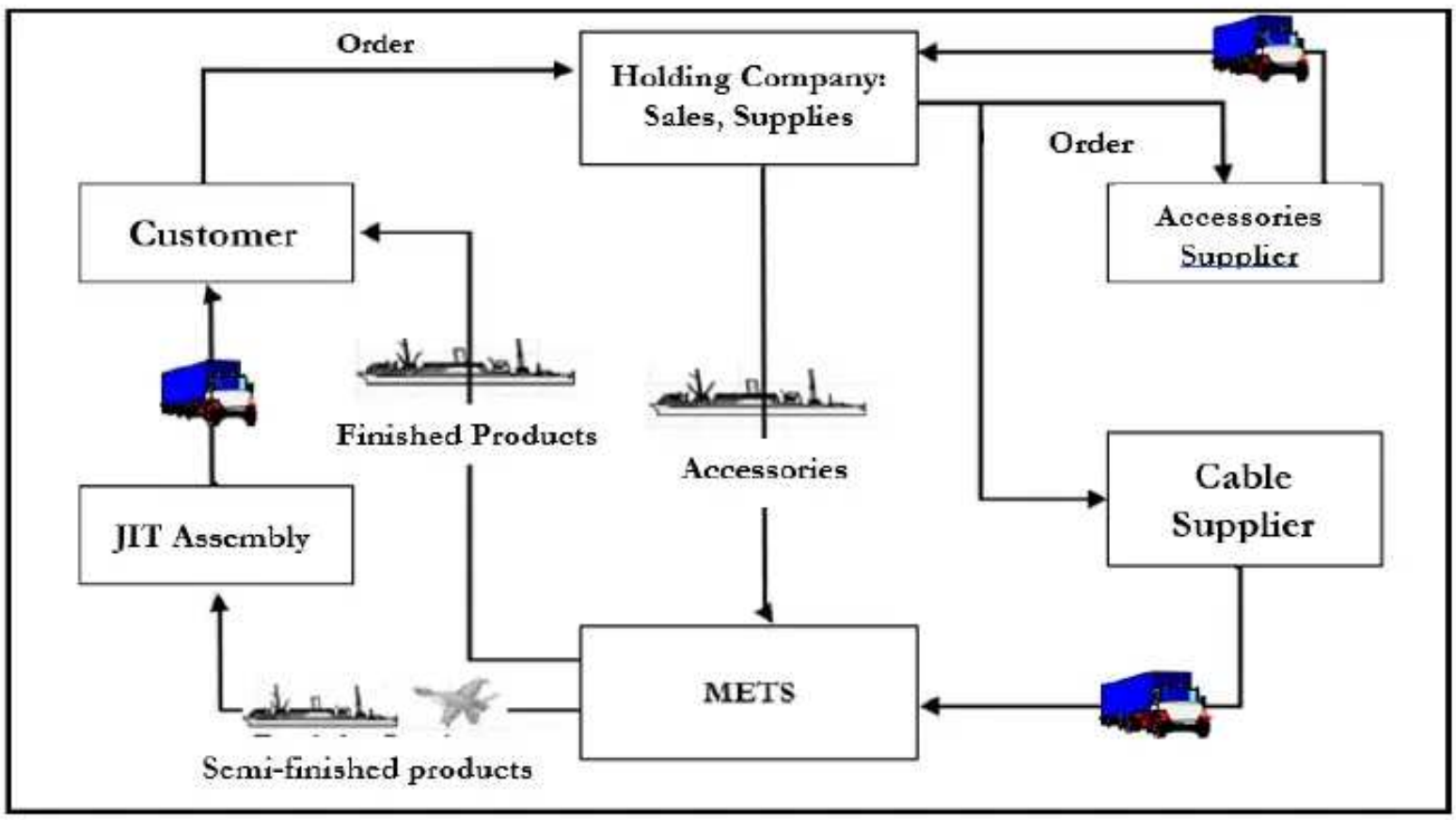


Figure 6: Order Flow: Customer Supplier

b. Data Flow

METS Data Flow involves several actors from reception of sales order until its supply (**Figure 7**); FST (Production Control) department is the key coordinator between FST of Vilsbiburg and METS departments that make the use of 2 principal platforms:

- KS-Sort: a management platform of TPM (Production Management Techniques) planned orders,
- KS-Info: dedicated to tools reservation for specific machines.

These platforms are exchanging data to facilitate production management and logistic process control. Feedbacks are sent in return if a problem occurs or just to give an actual report.

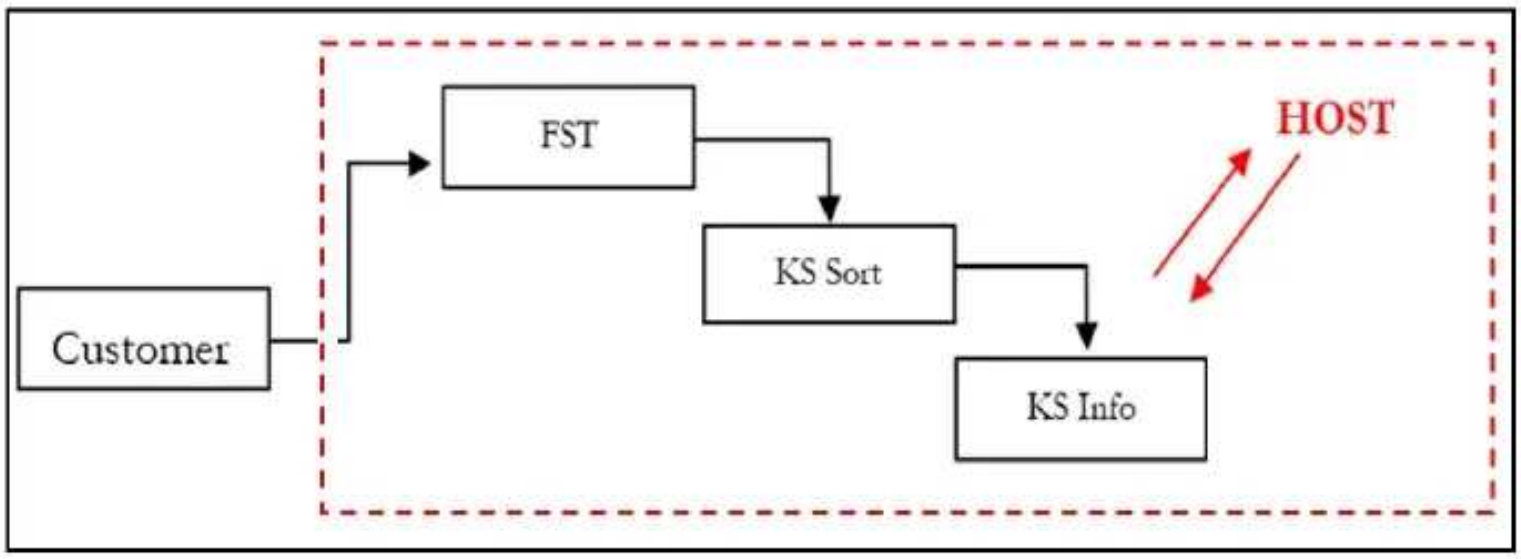


Figure 7: METS Data Flow Diagram

c. Materials Flow

KBE Electrotechnik is METS official supplier of raw material. It provides its customer with coils of normal and special cable wires stored in 07 cable Warehouse (**Figure 8**).

METS adopts a push-pull manufacturing strategy. Its activities are planned based on customer forecasts' need and specific functionalities. Orders planification includes reservation of:

- Accessories (Electrical connector, Tülles, special cable…),
- Coils of cable wires,
- Crimp Applicators.

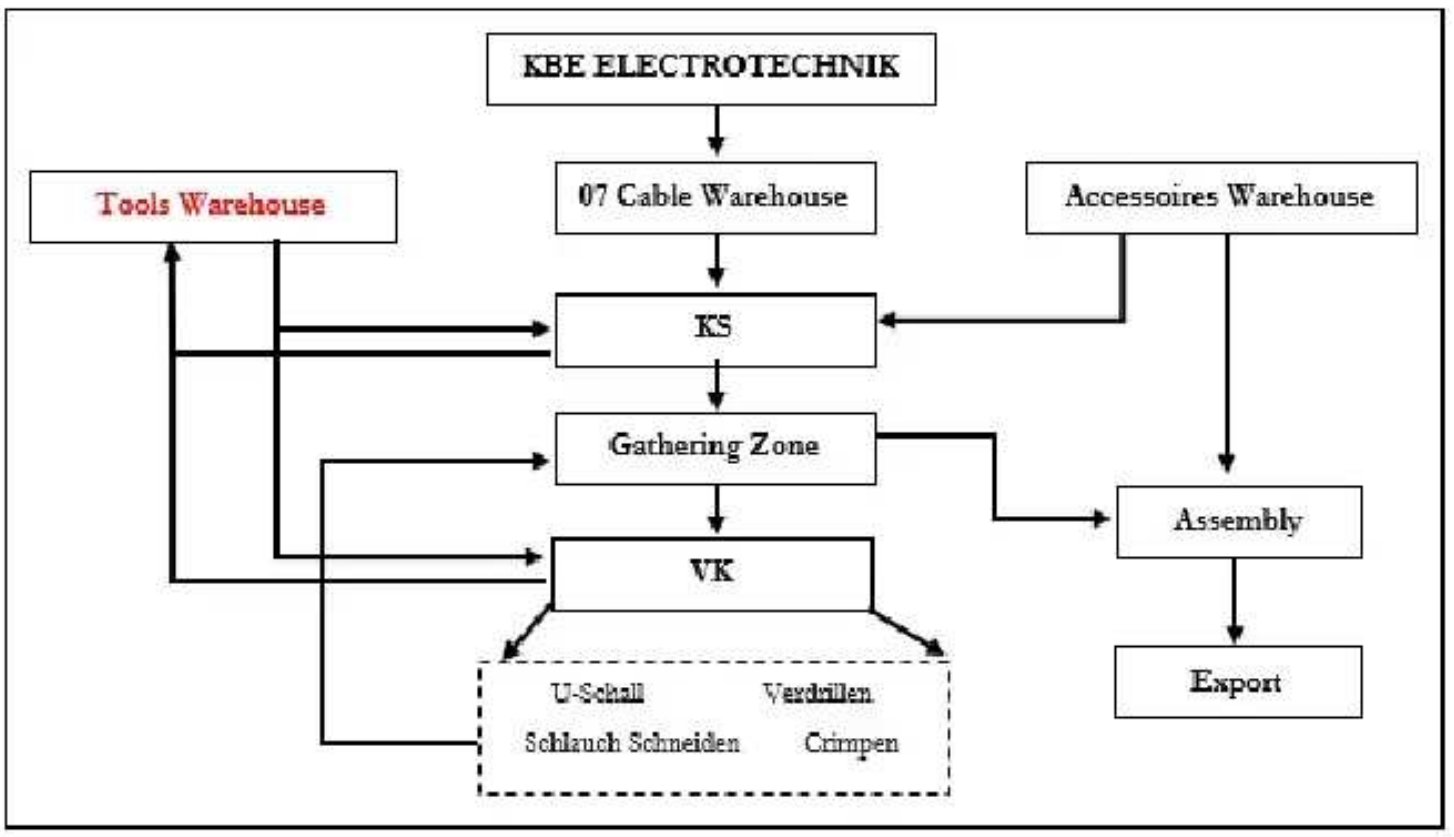


Figure 8: METS Materials Flow Diagram

2. Project Presentation

This part is dedicated to specifying the general context of the project, project requirements specification and the adopted approach.

2.1. Project General Context

Good logistics management is one of the most important factors for company's success. It has become a crucial element of competitiveness and therefore a condition improvement and success. For that reason, METS must set the record straight on good management of Tools warehouse flows for better control of tools inputs (storage) and outputs (retrieval). Catia 3D Drawing presented in **Figure 9** gives an overview of the current layout of tools warehouse.

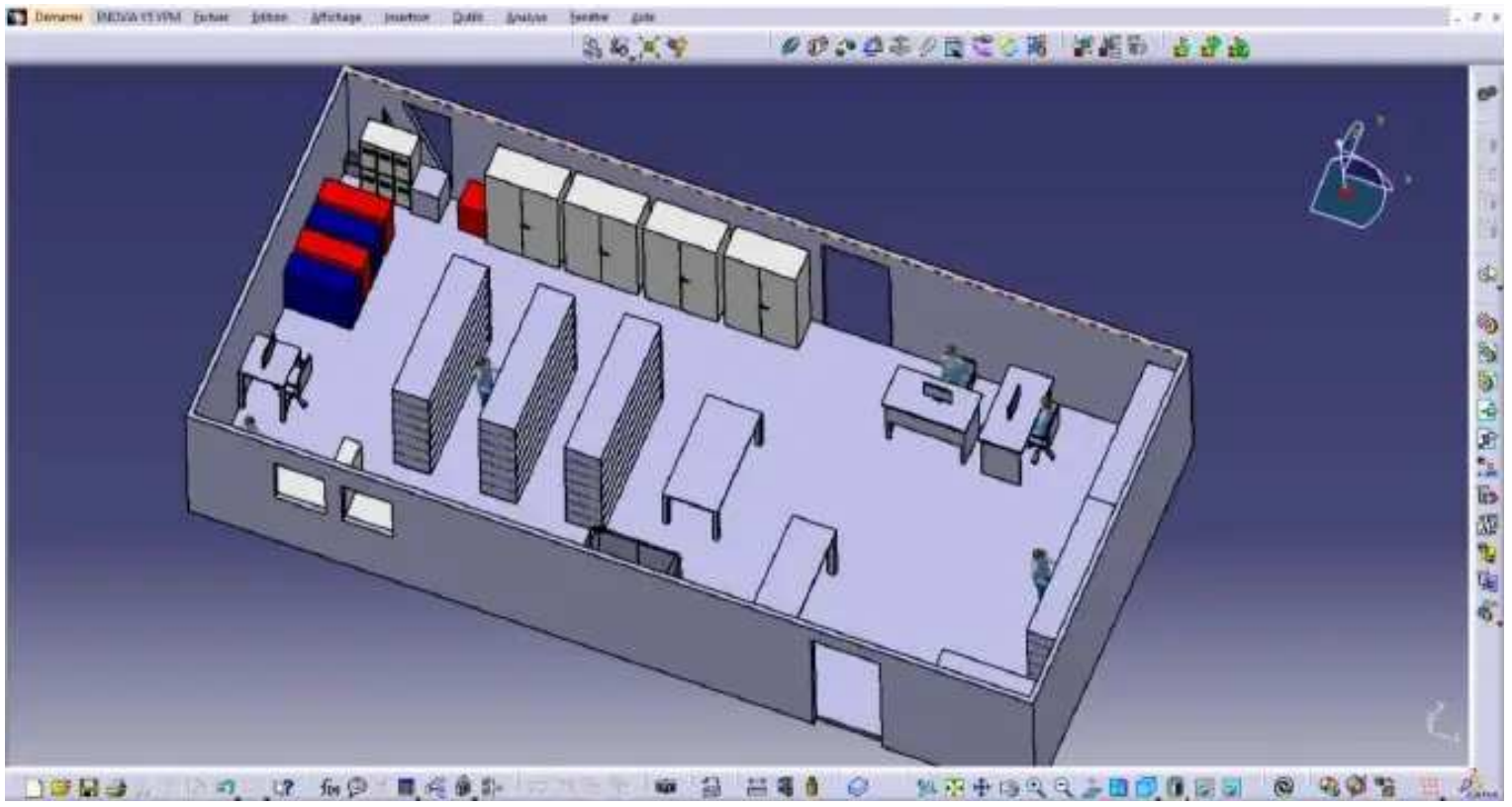


Figure 9: Tools Warehouse CatiaV5 3D Drawing

2.2. Project Requirements Specification

Maintenance KS-VK department is looking to improve the storage management of Crimp Applicators at Tools Warehouse and opts therefore for implementing an automated solution; "Automated Stacker crane" because of rapidly increasing number of crimp applicators and the deficiencies of current storage system.

Project requirements specification is a feasibility study which consists in:

- Designing a mechanical system able to securely manage the storage of **1300** crimp Applicators and reducing the current storage area by **20%**,

- Automating the proposed solution to increase warehousemen availability for repair and maintenance by **80%** and eliminate non-ergonomic work,
- Developing a Human-Machine Interface to facilitate system controlling and eliminating manual management errors.

2.3. Adopted Approach

Leading a project starts with choosing the approach and the methodology to organize the actions and define means to achieve goals.

DMAIC, the chosen approach, is one of the fundamental tools of Six Sigma. It is a proven problem-solving and a data-driven approach to improving processes in a logical and methodical way. It is designed to take a team through a process improvement project from inception to completion [3]. The five phases of DMAIC are:

- DEFINE: The purpose of this phase is to describe the problem, the project focus and the expected business benefits. One of its frequently used tools is the "Project Charter".
- MEASURE: The purpose of the second stage is to collect the necessary information to treat objectively the project and identify the critical process outputs.
- ANALYZE: it aims to list and prioritize potential causes of the problem.
- INNOVATE or IMPROVE: This step goal is to identify, test and implement a solution to the problem.
- CONTROL: consists in sustaining the gain by monitoring improvement to ensure continued success.

2.4. Project Charter

It's one of DMAIC first stage tools. It's a high-level document, the purpose executive of which is to get the project "green-lighted" by the executive sponsor and board of directors.

It outlines the project objectives, business case, stockholders involved in the project and a description of how the project will be organized and conducted, including start and end times, budget information and final deliverables [4].

Project Charter (**Table 2**) describes the problem in METS tools Warehouse and identify critical needs for customer: KS-VK production. It details risks and constraints related to the project as well as success factors and necessary budget.

To guarantee manageable project, to achieve fixed goals and to ensure that the deadline will be met, it is essential to set a work schedule based on the five stages of DMAIC approach.

Table 2: Project Charter

Project Charter

Project Name	**Mechanical and Automation of Engineering of Stacker Crane for Crimp Applicators**		
Responsible	**Amani GUEFRECH**	Accountable	**Mr. Younes RGUEZ**
Consulted	**Mr. Raouf KETATA Mr. Atef BOULILA**	Informed	**Mr. Wissem SASSI**
Starting Date	**02/01/2018**	Ending Date	**05/31/2018**

Problem Statement

Currently, Crimp Applicators Storage Management is no longer effective because of rapidly increasing number of tools at METS tools Warehouse.

Project Purpose(s)	• **Conception of Flexible Storage System**
	• **Implement new standard operating procedures**
	• **Reducing storage area by 20%**
	• **Increasing warehousemen availability for maintenance and repair at 80%**
Monitoring Indicators	• **Manuel Storage Rate**
	• **Operating time**
Project Boundaries	• **Variability of warehouse state (number of crimp applicators, change of layout)**
	• **DB limited Access**
	• **Necessity of manual storage stages**
Success Factors	• **Stakeholders availability**
	• **Coaching and Supervision**
Budget	• **Not fixed yet**
Risks	• **Overcrowded Storage Area**
	• **Increasing unavailability of technicians**
	• **Increasing risk Management and storage error**

DMAIC Planification

	Week																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Define		■	■															
Measure				■	■													
Analyze						■	■	■										
Innovate									■	■	■	■	■	■	■	■	■	
Control																		■

Schedule setting is based on the preliminary time estimation for the Innovative phase. DMAIC fourth phase takes time and needs resources allocation to be reached. So, it is necessary to take into consideration project conditions before setting the schedule: integration period and resources availability.

Project charter commits the working group in terms of deadlines which are detailed in the Gantt diagram (Appendix 1), in terms of expected results and underlines all the actors involved in this project and which are chosen based on RACI Matrix. It is the acronym of **R**esponsible, **A**ccountable, **C**onsulted and **I**nformed (**Figure 10**).

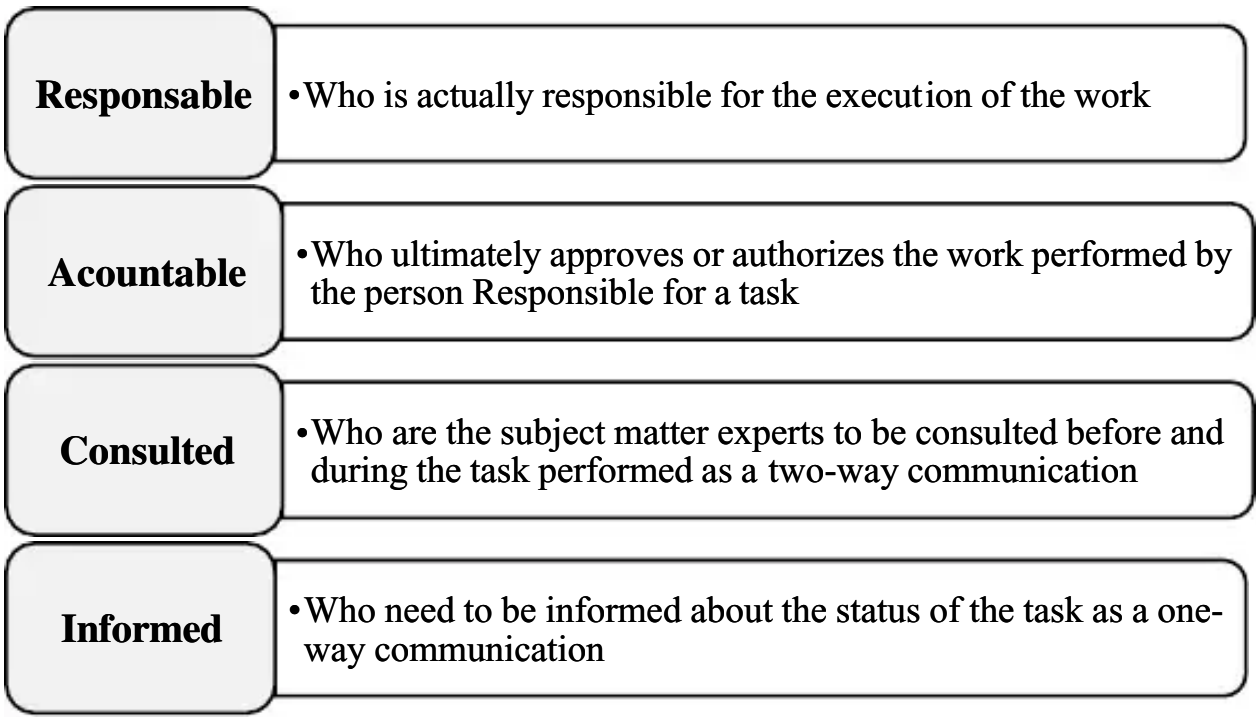


Figure 10: RACI Categories

Conclusion

In the first chapter, I gave an overview about my internship by presenting, the hosting company in the beginning then the project I had worked on and finally specifying the adopted approach. In the next chapter I will explain DMAIC second and third stages.

Chapter 2: Current State Analysis and Proposed Solution

Introduction

This chapter describes the current storage methodology. It introduces at first tool's warehouse, explains the deficiencies storage and management system and finishes by analyzing the proposed solution.

1. Current Storage Methodology

The first part of this chapter presents the tools warehouse and its characteristics.

1.1. Tools Warehouse

DRX. Tools Warehouse, located at KS-VK production site, occupies **102,374 m²** of surface area. It is Production supplier of Terminal Crimp Applicators (**Figure 11.a**) and Seal Modules which apply seals to individual conductor (**Figure 11.b**).

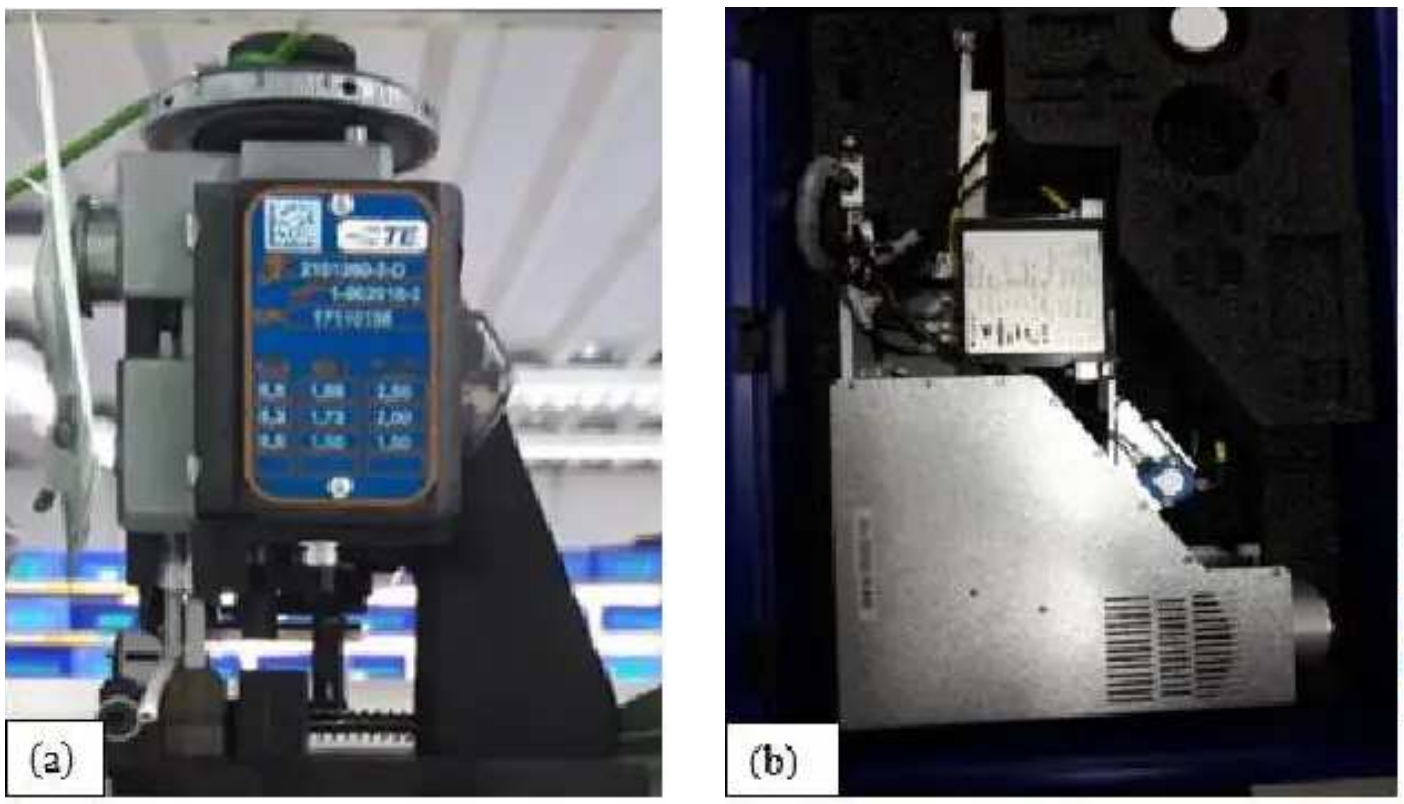


Figure 11: Managed Tools

1.2. Crimp Applicator's Specifications

These tools are referenced and stored in racks for better organization and minimum of storage area. Tools' allocated space width is about **180mm**: it facilitates handling by warehouse workers therefore avoiding damage to Crimp Applicators.

These tools can get different status depending on the allocated production site (VK or KS) and its technical status classified in the **Table 3**.

KS production is totally automated, and the allocation of crimp applicators is automatically managed through KS-Info platform while VK production is semi-automated and tools allocation is manually ensured.

Table 3: Comparison Between KS and VK Status

	KS Tools	VK Tools
Ready	10	05
Allocated	20	
In production	30	35
Repair	99	99

Different ranges of Crimp Applicators are available at tools warehouse (Tyco AMP, TYCO HDI, TYCO OCEAN, SCHÄFER, HANKE, SAW, DELPHI, SCHLEUNIGER). They present different weights and dimensions (**Table 4**) and each one of them is dedicated to crimping a specific cable which is detailed at tools cart which is among the highly confidential documents to the firm.

Table 4: Detailed Tools Characteristics

	Min Value	Max Value
Weight (kg)	3	14,5
Height Z (mm)	140	230
Width X (mm)	80	150
Length Y (mm)	150	340

Currently, the number of Crimp Applicators is rapidly increasing because of new project's need which is one of the study constraints. In **04/19/2018**, it reached **1216** tools with an increasing rate of **31%** compared to **03/01/2018.** To guarantee reliable results, it was necessary to ask for the approximative number of tools at the end of 2018 which is **1300** tools.

1.3. Storage System

1.3.1. System Standards' Description

Crimp Applicators storage area occupies **15%** of warehouse total surface. They are stored in Separate Racks formed by **8** shelfs with maximum storage capacity of 8 tools in each one. Current Storage System respects work instructions (**Appendix 2**) and 6S standards (5S standards with step of safety). 6S comes from 5 Japanese words:

- The first S: "Seiri" (Sort, Cleaning, Classify in English). Removing all clutter and unwanted items that are not in use.

The second S: "Seiton" (Straighten, Simplify, set in order, Configure in English). Using ergonomic principles to locate items where they minimize movement and reduce stress. Tools are straightened in ascending order referenced racks to facilitate storage management (**Figure 12**).



Figure 12: Referenced Racks

- The third S: "Seiso" (Clean and Check in English). This is the cleaning and painting of the area to bring it to a "new" standard.
- The fourth S: "Seiketsu" (Develop a discipline for SORT and SHINE in English).
- The fifth S: "Shitsuke" (To create a culture that daily follows these steps in English).

- The last S: “Safety”: additional focus to eliminate hazardous situations (**Figure 13**). Security equipment and signalization are therefore used to prevent risks.



Figure 13: Safety and Security Equipment’s

1.3.2. Risk Analysis at Tools ‘Warehouse

Risk analysis consists in assimilating various risks generated during the current management at tools’ warehouse. It is based on going to “Gemba”: it means to go and see what is really happening rather than talk about it or read about it.

In order to appreciate the detected findings, a scale is proposed distinguishing nine levels of risks which facilitate problem location and therefore corrective decision-making.

The risk management is based on risk level to prioritize and to reduce risk impact and appearance probability. The risks in the red box presented in the **Table 5** are the most critical ones.

Table 5: Risk Assessment at Tools' Warehouse

Impact				
	Significant		**Non-Ergonomic Storage System**	**Technicians Unavailability for storage**
	Moderate	Tools' accident	**Manual Management Errors**	**Overcrowded Warehouse**
	Minor		Inflexibility of current Storage System	
		Low	Medium	High
		Probability		

1.3.3. System Deficiencies

Risks in Tools' warehouse are the result of management system deficiencies whose causes are detected with reference to technicians' questionnaire and visuals analysis:

a. 5M or Ichikawa Diagram

5M identification allows to detail different causes of the problem which are due to Manpower, Machines, Methods, Mother Nature and Materials:

Man Power or physical work: causes due to the staff working in warehouse:

- Only 2 Technicians responsible for tools management at the warehouse (storage, change of status, repair, maintenance),
- Lack of technician's motivation,
- Some Storage standards are not respected (illegible reference (**Figure 14**), hand-written reference (**Figure 15**), non-respect of storage decrescent order (**Figure 16**) ...).

Figure 14: Illegible Reference

Figure 15: Hand-Written Reference

Figure 16: Non-Respect of Storage Decrescent Order

Machines: causes due to tools and equipment used to complete production task and facilitate physical work. In tools' warehouse they do not exist: Lack of handling system.

Methods: causes due to production and support processes

- Separate storage areas (additional racks in case of additional tools)
- Expensive and Weak Tools Label (they can't resist more than 2months) (**Figure 17**)



Figure 17: Tools defected Label

- Repair and maintenance zone is generally overcrowded of Crimp Applicators because of unavailability of technicians (**Figure 18**)



Figure 18: Repair and Maintenance Area

Environment: causes due to environmental system

- Limited storage surface

Materials causes due to components and general materials used for business operations (Crimp Applicators)

- Diversity of dimensions
- Numerous Tools (about 1200 Crimp Applicator)
- Different width (from 3 to 14,5kg)

The diagram bellow is the graphical representation of different causes found called “Ichikawa Diagram” it summarizes the sources of problems detailed before (**Figure 19**)

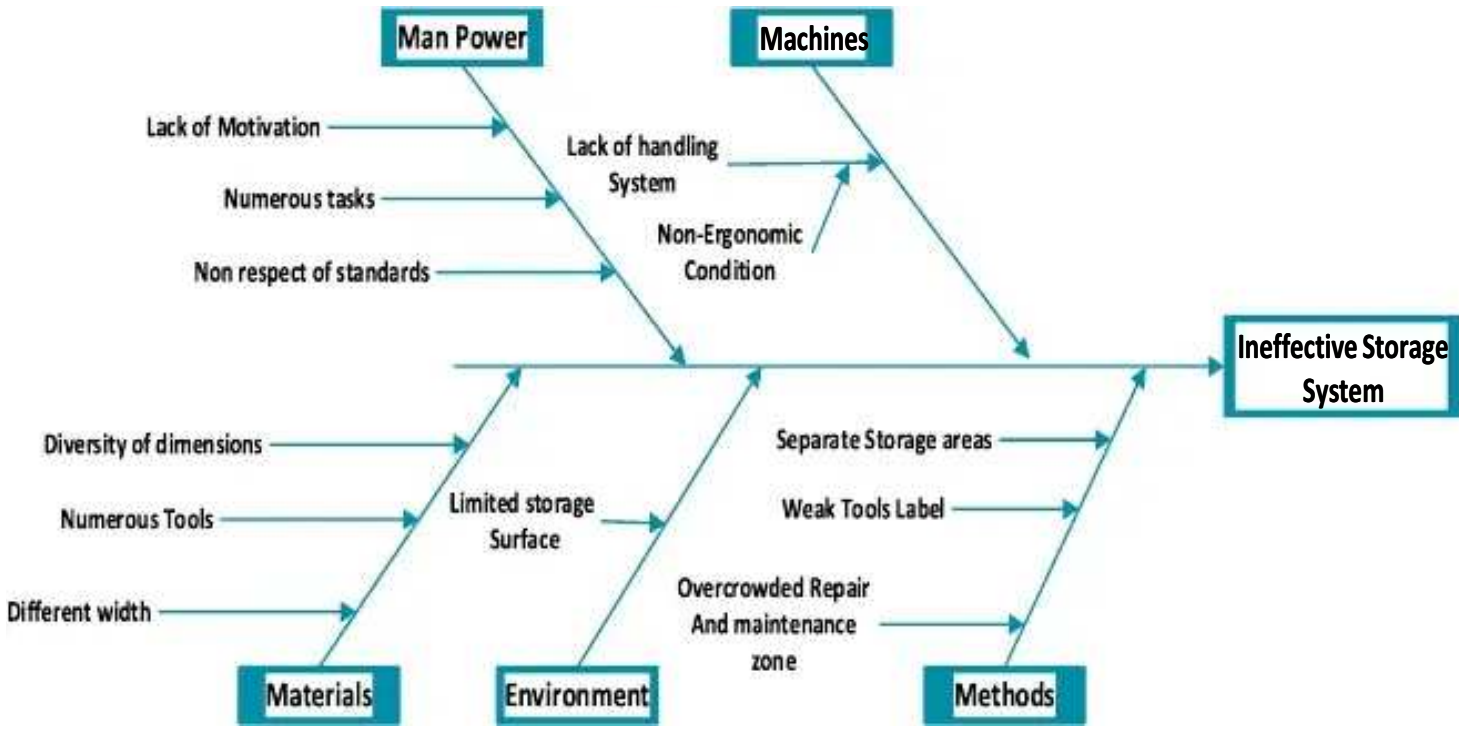


Figure 19: Ichikawa Diagram

b. Ergonomic Analysis

Non-ergonomic condition is one of the causes leading to an ineffective storage system as mentioned in the Figure 19. RULA method is used to evaluate the work posture and analyzing the non-ergonomics. It is associated with the upper limbs of the body, presenting fatigue essential sources for the case of study.

It makes possible to highlight the risk of the postures related to the work cycle observed and to calculate the score to decide the corresponding improvement actions and the level of urgency of their implementation.

Catia V5 software is used to calculate the RULA score (**Figure 20 and Figure21**) corresponding to the most critical posture at tool storage (or retrieval) from (or on) rack’s last shelf. RULA final score is detailed as bellow in **Table 6**:

Table 6: RULA Scoring

Scoring	Score Color	Action / Assessment
1 or 2		Acceptable posture
3 or 4		Further investigation, change may be needed
5 or 6		Further investigation changes soon
7		Investigate and implement change



Figure 20: RULA score for 1st Critical Posture

According to RULA assessment (**Appendix 3**), final score of the study case reached 6 (**Figure 20**). This posture presents a medium risk therefore it requires investigation and rapid changes.



Figure 21: RULA score for 2nd Critical Posture

According to RULA analysis (**Appendix 3**), final score of the study case reached 7 (**Figure 21**). This posture presents a very high risk therefore it requires immediate actions.

2. Proposed Solution

To limit the risks and deficiencies of current storage system, implementing an automated management is one of the suitable proposed solutions.

2.1. Automated Stacker Crane

Logistics systems and processes are opting for modern technologies because of their numerous advantages:

- Improving the efficiency of storage,
- Facilitating handling and preparation of tools, raw materials and goods,
- Eliminating manual error,
- Assuring human safety,
- Traceability.

Stacker Crane is one of these modern technologies. It is a totally automatic machine. Its movements respond to storage and extraction orders sent by a control system that optimizes these movements [5].

In this case of study, the solution proposed consists in designing and automating a trilateral stacker crane for crimp applicators. It insures three principal functions: translation X axis, lifting Z axis and gripping Y axis.

2.2. Horned Beast Flowchart

The diagram in **Figure 22** defines the need to which the stacker crane answers:

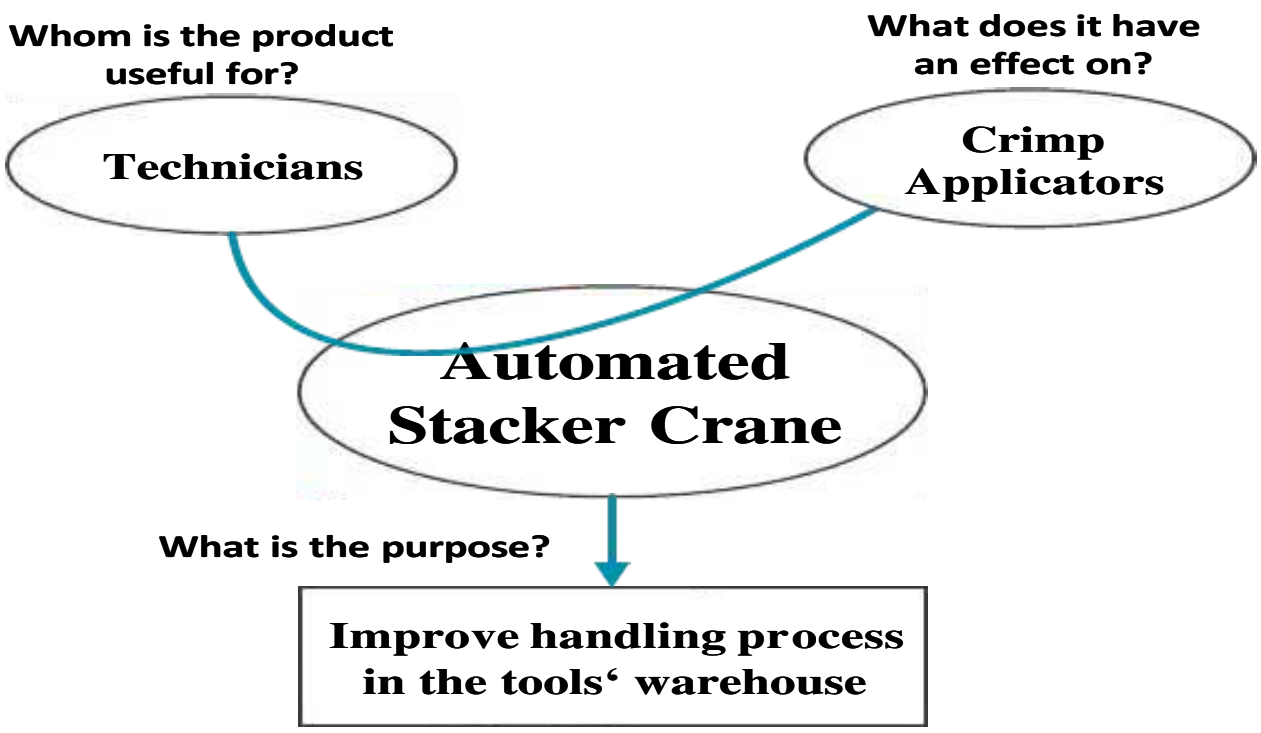


Figure 22: Horned Beast Diagram

2.3. Octopus Diagram

The diagram bellow (**Figure 23**) defines the principal functions and constraints in the environment:

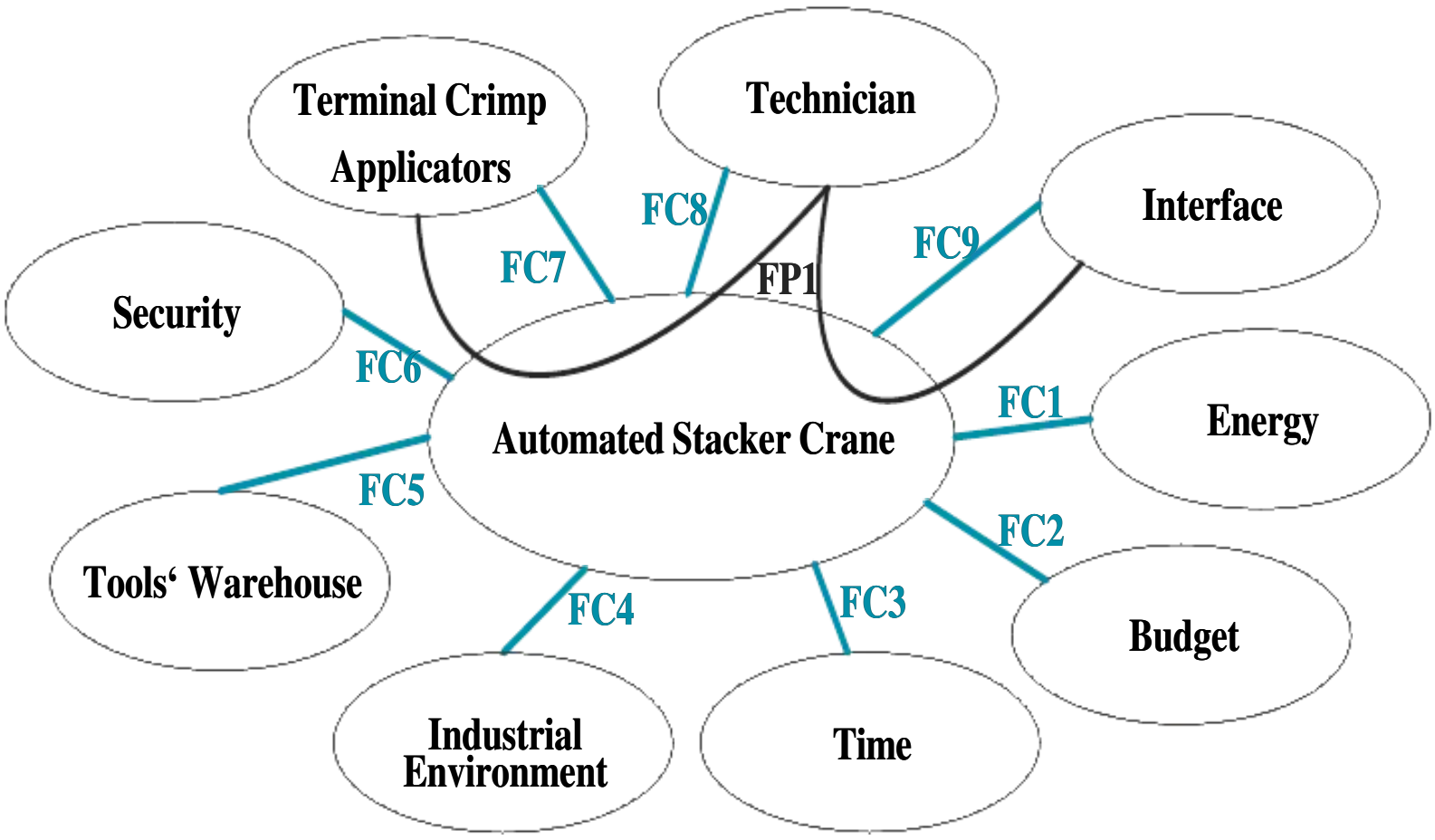


Figure 23: Octopus Diagram

Principal and constraint functions are detailed bellow:

FP1: To be able to control crimp applicators handling through an interface.

FC1: To connect to the electrical network

FC2: To minimize the budget
FC3: To optimize handling time

FC4: To be robust regarding the environment

FC5: To respect Warehouse lay-out

FC6: To guarantee the safety and security of technicians and tools

FC7: To be adapted to tools dimensions and geometry

FC8: To ensure the technicians ergonomics

FC9: To be adapted to crimp applicators storage system.

To prioritize and evaluate Stacker Crane service functions, a brainstorming took place with the project accountable to set a paired comparison Matrix illustrated in **Table 7**. It consists in attributing a weight to each function or a value related to its importance:

Table 7: Paired Comparison Matrix

	FC1	FC2	FC3	FC4	FC5	FC6	FC7	FC8	FC9	NOTE	%
FP1	FP1.**2**	0	FC3.**2**	0	FP1.**3**	FC6.**2**	0	FC8.**2**	FP1.**2**	7	**8%**
	FC1	FC2.**3**	FC3.**3**	FC1.**1**	FC1.**2**	FC6.**3**	FC7.**3**	FC8.**2**	FC9.**1**	2	**2%**
		FC2	FC3.**1**	FC2.**2**	FC2.**3**	FC6.**2**	FC7.**1**	FC2.**1**	FC2.**2**	11	**13%**
			FC3	FC3.**2**	FC3.**3**	FC6.**2**	FC7.**1**	FC8.**1**	FC3.**3**	14	**16%**
				FC4	FC4.**3**	FC6.**3**	FC7.**3**	FC8.**3**	FC4.**2**	5	**6%**
					FC5	FC6.**2**	FC7.**1**	FC8.**3**	FC5.**1**	1	**1%**
						FC6	FC6.**2**	FC6.**1**	FC6.**3**	20	**23%**
							FC7	FC8.**1**	FC7.**2**	11	**13%**
								FC8	FC8.**3**	15	**17%**
									FC9	1	**1%**
									Total	87	100%

Scale: 0: Equal 1: Slightly higher 2: Moderately higher 3: Exclusively Better

Using the table above, Pareto diagram (**Figure 24**) also known as the 80-20 rule, has been designed to identify the most critical service functions:

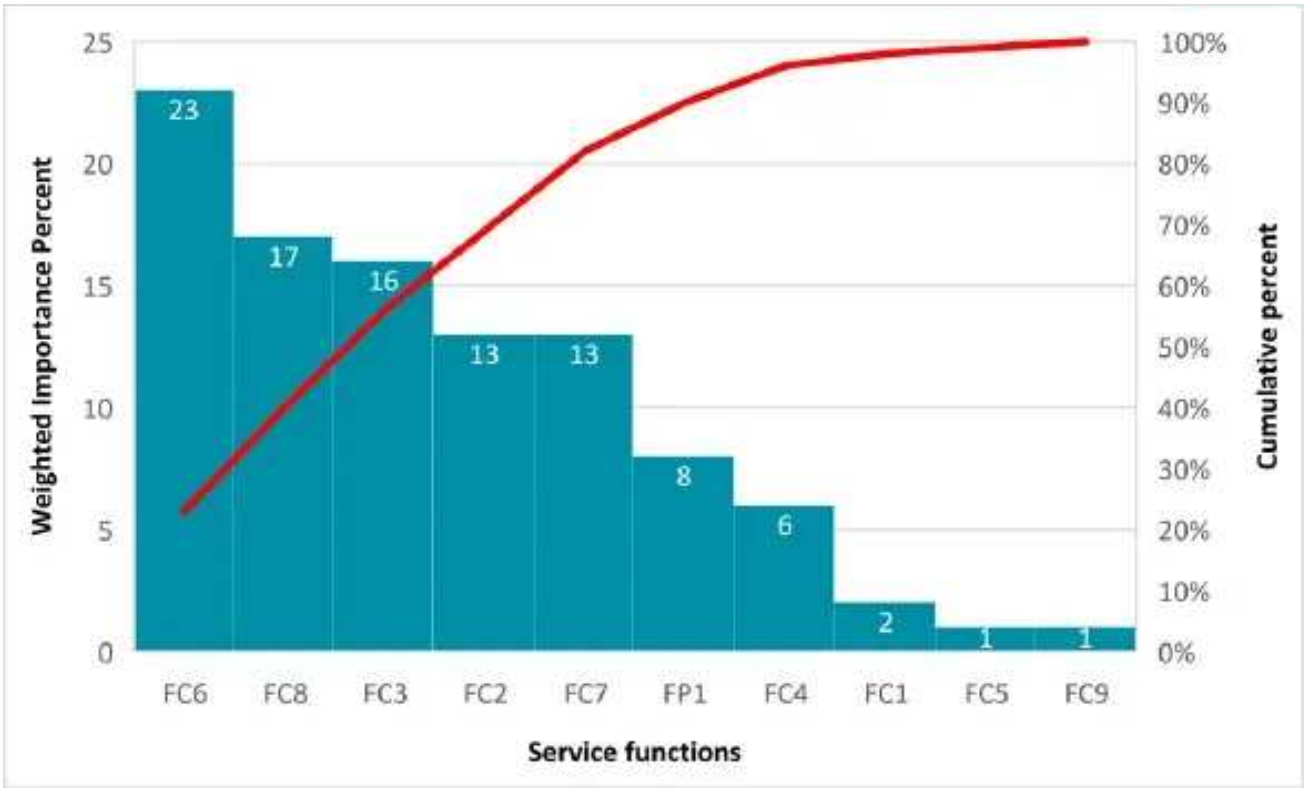


Figure 24: Pareto Diagram

40% of total service functions, which are: Security, Ergonomics, Time and Budget, are priority ones to take into consideration while choosing the technological solutions.

2.4. FAST Analysis

Function Analysis System Technique is a value engineering phase. The **Figure 25** illustrates the Stacker Crane FAST. It is based on Octopus Diagram to make the right choice of technological solution by taking into consideration the four priority constraint functions: Security, Ergonomics, Time and Budget.

The principal function needs at first:

- The control of storage and retrieval of crimp Applicators which is ensured by a Programmable Logic Controller. The choice of this solution is due to numerous advantages (**Table7**): the safety and industrial grade (Tolerance to withstand various hazardous condition for industries such as Electromagnetic interference, high temperature, power fluctuation…), easy maintenance (Due to modular design the trouble shooting is easier), complexity, integration, reliability…

Table 8: Comparison between PLC and Microcontroller

	PLC	Microcontroller
Application	**Industrial controlling machinery**	**Different types of applications**
Robustness	**Immunity to electronic noise and vibration Extended temperature range.**	Weak Noise and Vibration immunity.
Modularity	**Yes**	No
Cost	High	**Low**
Programming language	**Easy**	Complex
Memory	**increasable**	Limited
Watchdog timers	**increasable**	Limited
Counters	**increasable**	Limited
Size	Bulky	**Smaller**
I/O	**Infinitely Expandable**	Finitely Expandable

- The Vertical Movement which is ensured by Rack and Pinion. In comparison with “belt and pulley” and “screw nut”, “rack and pinion” is much less likely to experience catastrophic failure in vertical applications. This mechanical system is consisting of a rack, a pinion and a gearbox. The gearbox contributes to optimizing the speed of the

motor and therefore adapting the torque needed. Belt driven actuators aren't recommended by manufacturers in vertical applications, due to the rick of damage if the belt breaks or the brake mechanism fails. [6]

- The horizontal Movement which is ensured by Belt and Pulley. This choice is based on the long travel length "eight meters for horizontal translation". The **Table 8** illustrates the comparison between the three different movement transformation mechanisms:

Table 9: Comparison between movement Transformation mechanism

	Screw +nut	Pinion + rack	Pulley Belt
Performance	Good	Good	Very Good
Reversibility	Yes	Yes	Yes
Friction	Low	Moderate	Moderate
Speed reduction	Very important	Important	Important
Stroke	Limited	Limited	Non-limited
Travel speed	Moderate	High	High
Cost	High	Moderate	Moderate
Inertia	Low	Moderate	Moderate
Transported charge	Moderate	Important	Low

Conclusion

In this chapter, I explained at first the motivation leading to the automated storage at tools-warehouse: the current system deficiencies then I analyzed the methodology adopted to choose the right solution. In the next chapter, I will detail the specificity of the mechanical and electrical solutions which is the second stage of the DMAIC innovate phase.

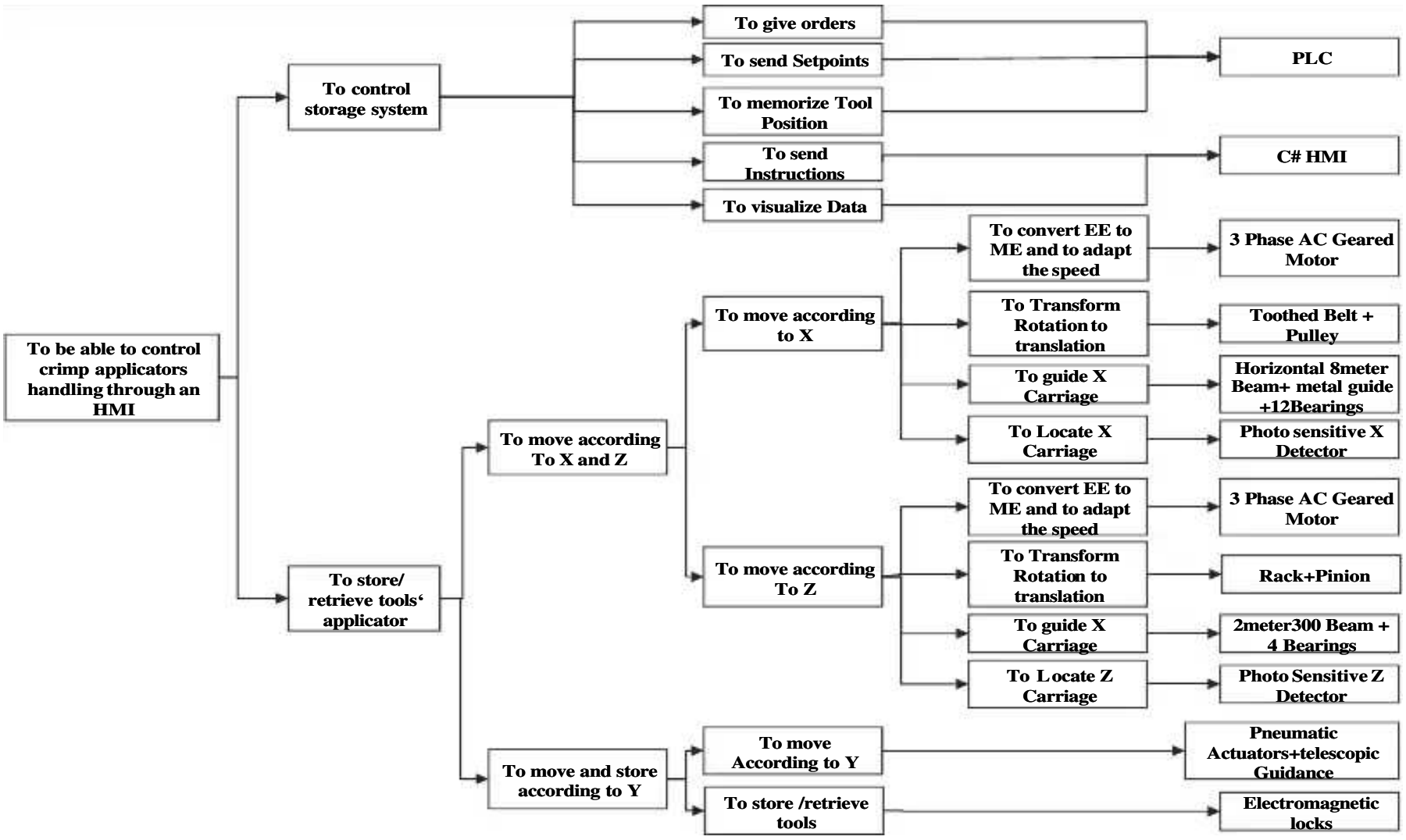


Figure 25: FAST Diagram

Chapter3: Mechanical and

Electrical Design

Introduction

In this chapter, the second stage of DMAIC Innovative phase is dedicated to detailing at first the mechanical solution using CATIA V5 a multiplatform software then presenting the corresponding sensors, actuators and automation dimensioning and finishing with the electrical wiring diagrams.

1. Mechanical Dimensioning

1.1. Technological Concept Diagram

The mechanical storage system, as mentioned in the second chapter, is constituted of three principal movements according to X, Z and Y directions. A general concept is presented in the Figure 26 introducing different mechanical parts. The appendix 4 shows the stacker crane 3D design.

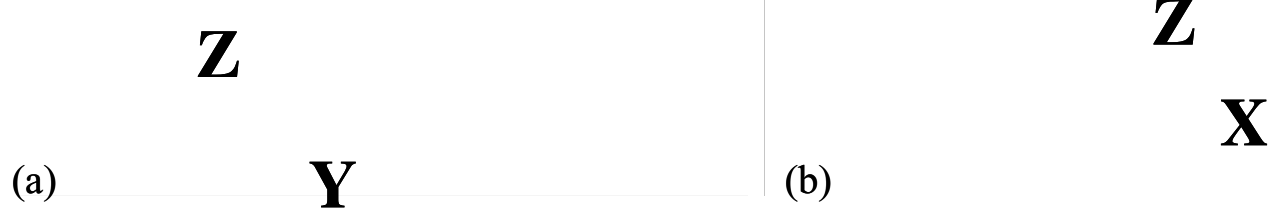


Figure 26: Stacker Crane Technological Concept

1.2. Storage Area

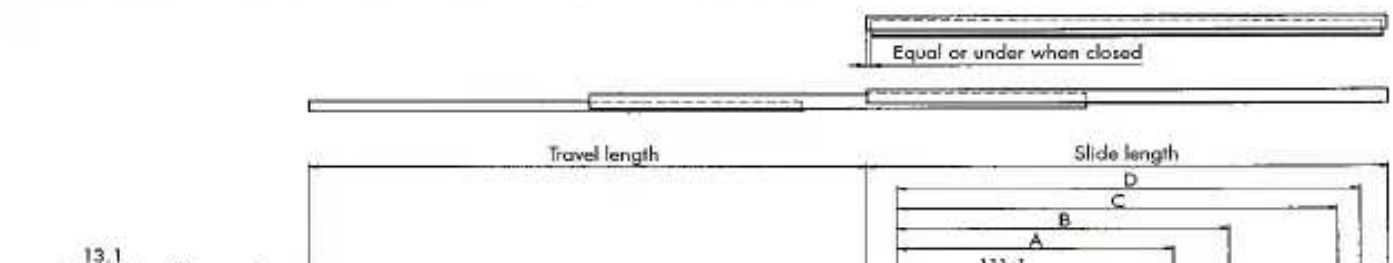
The stacker crane stores crimp applicators in two specific areas. Each one of them is constituted of 6 racks with (W x L x H) = (1300 x 360 x 2160) dimension. One rack enables to store 56 Tools (8 x 7). If we take into consideration the storage and retrieval space in both areas

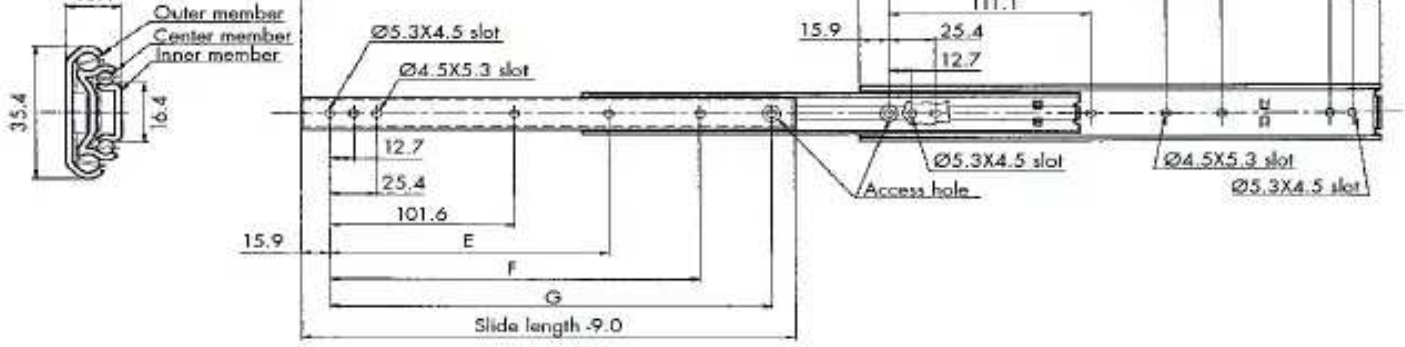
accessible to the operator (black colored spaces in Figure 26), total storage capacity is about 670 tools.

1.3. Retrieval Function (Y)

The movement according to the Y axis is assured by pneumatic actuator. It is associated to a telescopic electromagnet to across the depth of the racks extracting or positioning crimp applicator. The electromagnet is fixed to HTS telescopic ball bearing slides presented in the **Figure 27** assuring its translation. The choice is based on the maximum charge supported 29kg, travel performed 360mm, and safety guaranteed.

The supplier of TS-350 -over travel compact slide- is HepcoMotion. It provides a variety of dimensions depending on customer technical needs: the travel length that matches tools location depth is 360mm.





Part number	Slide length	Travel length	Hole length mm							Load Rating kg/pair	Slide weight kg/pair
			A	B	C	D	E	F	G		
TS–350–8	203.2	214.0	–	–	158.7	171.4	–	50.8	152.4	33	0.50
TS–350–10	254.0	265.0	–	–	209.5	222.2	–	101.6	203.2	32	0.64
TS–350–12	304.8	316.0	–	149.2	260.3	273.0	–	152.4	254.0	30	0.73
TS–350–14	355.6	366.0	–	200.0	311.1	323.8	–	203.2	304.8	29	0.85
TS–350–16	406.4	417.0	–	250.8	361.9	374.6	–	254.0	355.6	28	0.98
TS–350–18	457.2	468.0	212.7	301.6	412.7	425.4	207.7	304.8	406.4	26	1.11
TS–350–20	508.0	519.0	238.1	352.4	463.5	476.2	233.1	355.6	457.2	25	1.23
TS–350–22	558.8	570.0	263.5	403.2	514.3	527.0	258.5	406.4	508.0	24	1.35
TS–350–24	609.6	620.0	288.9	454.0	565.1	577.8	283.9	457.2	558.8	23	1.46

Figure 27: HepcoMotion TS -350 proposed dimensions

Cylinder pivot bracket guarantees the association of pneumatic actuator and the HTS component and therefore ensuring the translation of the retrieval system.

1.4. Lifting Function (Z)

Vertical movement is assured by HDS2 components as shown in Figure 29. This model, supplied by Hepcomotion, is chosen to ensure securely the 2350mm translation. It consists of an Assembled Beam and Rack Driven Carriage.

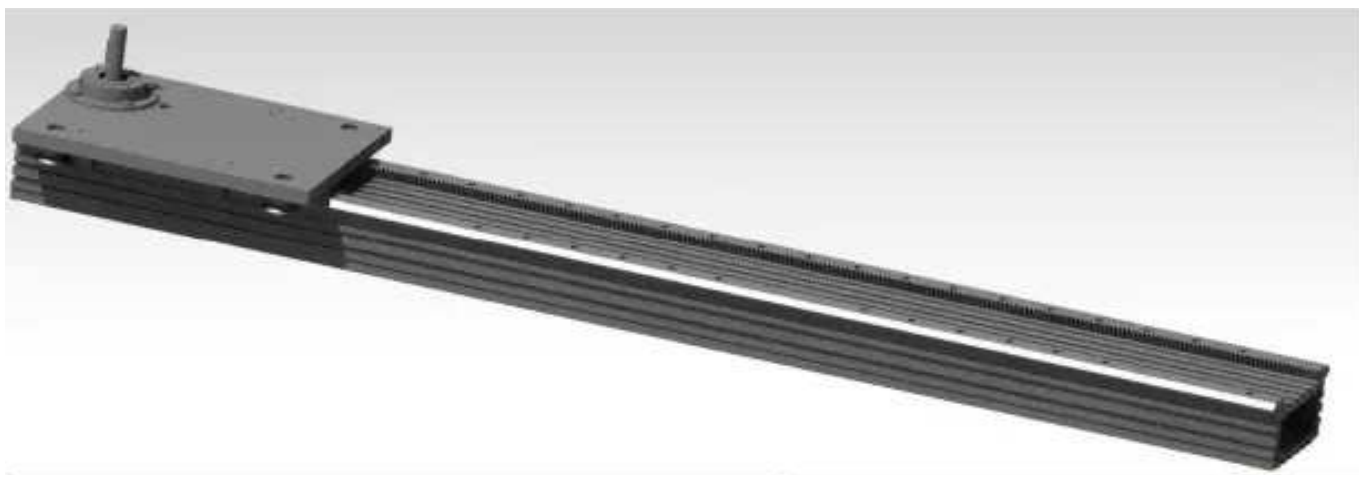


Figure 28: HDS2 Catia 3D Design

1.4.1. Assembled Beam

This assembled part consists of a high strength lightweight aluminum beam. The figure 29 specifies the width of HB 25 and its second moment of Inertia (Beam Deflection can be calculated using simple beam theory requiring this moment), it also indicates the single edge slide compatible with HB 25 and its width (24x2,3 = 55,2kg).

Beam	Single edge slide & flat track			
	HSS25	HTS25	HSS33	HTS33
HB 25C	✓	✓	×	×
HB 25	✓	✓	×	×
HB 33	✓*4	✓	✓	✓
✓ = Standard option × = Not compatible				

Beam	Second Moment Of Inertia		
	Vertical X-X	Horizontal Y-Y	
HB 25C	$2.8x10^6$	$10.2x10^6$	11.3kg/m
HB 25	$4.7x10^7$	$1.8x10^7$	24kg/m
HB 33	$16.9x10^7$	$8.4x10^7$	37.5kg/m
Beam second moment of inertia figures, are stated in mm^4			

Figure 29: Hepcomotion HB proposed specification

X-X(Ix) represents second moment of area (of inertia) with respect to the verical-axis; Y-Y(Iy) represents second moment of area with respect to the horizonal-axis. They are calculated with reference to beam dimensions to define resistance capacity or deflection. The Figure 30 presentes a simple verification of HB 25 Second Moment of Inertia using Catia V5:

IoxG=>Y-Y and IoyG=>X-X

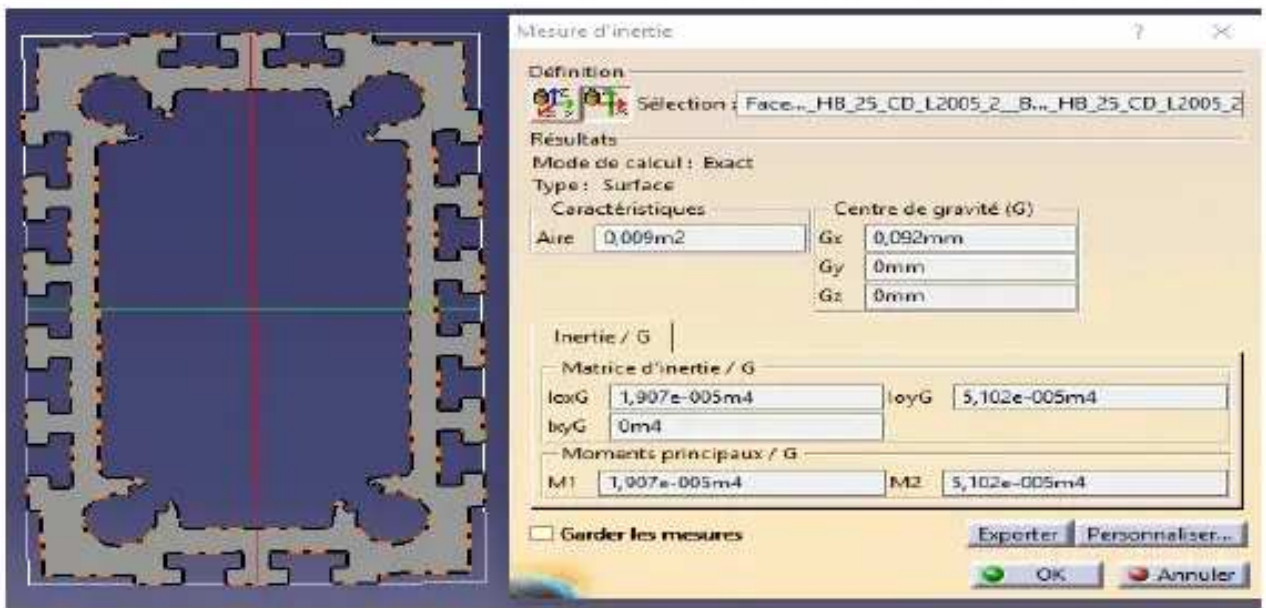


Figure 30: Catia V5 Inertia Measure

CHSS25 L2350 D57 is Hepco HDS2 V slides which is compatible with HB 25. It is chosen with a spur rack to garantee the charge vertical mouvement. The rack module is 2.5.

1.4.2. Rack Driven Carriage

The carriage (Figure 32) is dedicated to the construction beam assembly already chosen in the paragraph 1.4.1. The drive flange assembly is designed to mount on the top face of the rack

driven carriage and connect the drive from the geared motor, to the pinion. The dynamic radial bearing capacity of HDF25 is 38000N. The chosen pinion counts 20 teeth and has a diameter of 50mm corresponding to the referenced part HP 2.5 X 20.

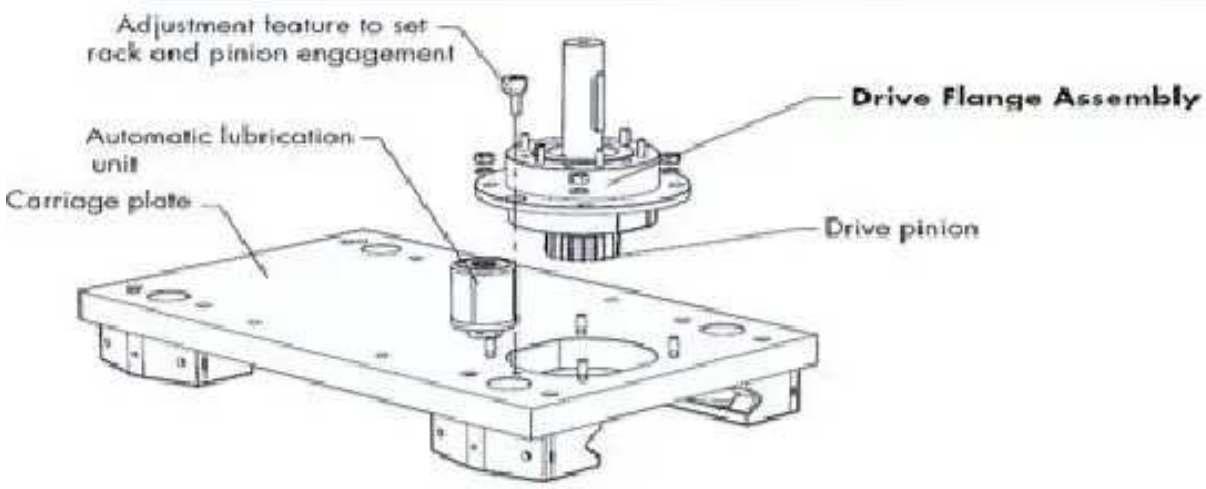


Figure 31: Rack Driven Carriage 3D design

1.5. Gripping Function (X)

Horizontal movement is assured by DLS4 components. This technology is chosen to ensure securely the 8000 mm translation. It is based on the toothed belt and pulley movement transmission. The carriage is fixed to the toothed belt with a terminal strip to fasten the free ends of belts as it is presented in Figure 32.

The model is equipped with V bearings facilitating the carriage translation. This movement is insured by a 100mm pulley diameter and a specific belt.

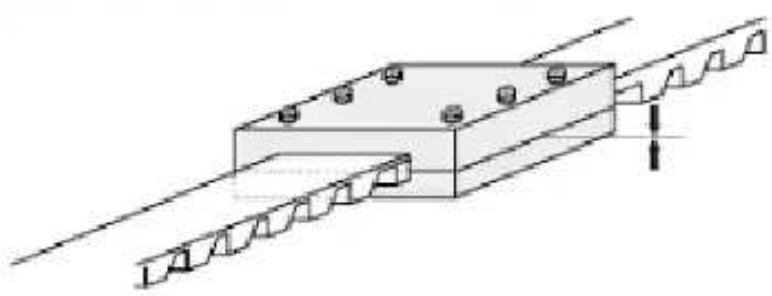


Figure 32: Terminal Strip Positionning

To specify Belt characteristics with reference to Forbo supplier [7], the table 11 illustrates symbols and designation of used parameters.

Table 10: Initial Parameters

Designation	Symbol	Value	Designation	Symbol	Value
Travel	S_v	7824 [mm]	**Mass of slide**	m_L	140[kg]
Speed	**v**	0,8[m.s^{-1}]	**Slide length**	**L**	116[mm]
Acceleration	**a**	0,2[m.s^{-2}]	**Pitch diameter**	d_0	100[mm]

			diameter		
Friction Coefficient	**μ**	0,5	**Acceleration due to gravity**	**g**	9,81[m.s^{-2}]
Operational Factor	**c_2**	1.4	**Acceleration Factor**	**c_3**	0
Teeth in mesh factor	**c_1**	12	**Aluminum Density**	**ρ**	2,7 [kg.dm^{-3}]
Permissible tension Member load	**F_{per}**	Table value from calculation sheet	**Outside Diameter**	**d_k**	(catalogue supplier)

Operational factor, teeth in mesh factor and acceleration factor are values of construction.

Equations below are used to specifying tooth belt dimensions:

Accelerating force F_A: $F_A = m_L . a$ $[N]$ (1)

Frictional force of guide rails: $F_R = m_L . g . \mu$ $[N]$ (2)

Speed (RPM): $n = \frac{v * 60 * 10^3}{D_0 * \Pi}$ $[min^{-1}]$ (3)

Effective pull to be transmitted: $F_u = F_A + F_R$ $[N]$ (4)

Maximum effective pull: $F_{umax} = (c_2 + c_3) . F_u$ $[N]$ (5)

Specific effective pull required: $F'_{ureq} = \frac{F_{umax}}{c_1}$ $[N]$ (6)

Toothed Belt selection:

(3): $n = 152{,}8\ tr/min^{-1}$

(4): $F_u = 725{,}9N$

(5): $F_{umax} = 1016{,}26N$

(6): $F'_{ureq} = 84{,}69N$

For linear drives, it is preferably to use AT and HTD belt. There are three possible types: AT 5, AT 10, HTD 8M.

The Appendix 5.a gives an overview of different types. The selection of the belt is based on the combination of (n, F'_{ureq}) and D_0. Therefore, the corresponding belt type is AT10 (high spring resistance, Belt pitch t=10).

The Appendix 5.b facilitates the definition of specific effective pull $F'_u = 160N$, belt width b_0=25 and $F_{max} = 3840N$. Therefore, the selected toothed belt is 25AT10.

S_0 25 and Y_{per} = 35 N/m². Therefore, the selected toothed belt is 25AT10.

Toothed belt Pulley selection:

Teeth Number:	$Z = \frac{D_0 . \Pi}{t}$	---	(7)
Mass of Timing belt pulley:	$m_Z = \frac{(d_k{}^2 - d^2)\pi\, b\, \rho}{4.10^6}$	$[N]$	(8)
Reduced Mass of Timing belt pulley:	$m_{Zred} = \frac{m_Z}{2}(1 + \frac{d^2}{d_k{}^2})$	$[N]$	(9)
Length belt untensioned:	$l = 2e + Z * t$	[m]	(10)
Mass of Belt:	$m_R = m'_R . l$	[kg]	(11)
Tooth base service factor:	$S_{teeth} = \frac{F'_u}{F'_{pureq}}$	---	(12)

(7): Z=31,4

The selected standard pulley and adapted to AT10 is the one with Z_p=32 teeth

$d_0 = \frac{Z_p . t}{\Pi}$ => $d_0 = 101{,}86mm$

$d_k = 100mm$; $d = 24mm$; $b = 32mm$

Therefore:

(3): n=150 tr/min

(8): $m_Z = 0{,}64kg$

(9): $m_{Zred} = 0{,}34kg$

(10): $l = 16{,}2\text{m}$

(11): $m_R = 2{,}6kg$

(12): $S_{teeth} = 1{,}89 > \mathbf{1}$

To summarize:

- Toothed Belt is 25AT10, 16,2m of length
- Toothed belt pulley is with Z=32 for 25mm belt and n=150tr/min

2. Instrumentation

This part is dedicated to the instrumentation of actuators, pre-actuators and sensors.

2.1. Actuators dimensioning

Three principal actuators are taking parts in the stacker crane functioning: Pneumatic Cylinder, Electromagnet and 2motors.

2.1.1. Pneumatic Cylinder specifications

For each shelving; left and right racks; two pneumatic cylinders are needed to drive the electromagnet to tools' storage box. This pneumatic component is obliquely fixed on the carriage to minimize the occupied spaces (122degre, to guarantee the penetration of 360mm of depth conforming to system requirements and securely push and pull maximum load (14,5kg $+M_{electromagnet}$)). Three criteria contribute to choosing the pneumatic cylinder Figure 33:

- The course: 250mm
- The direction: Double
- Radial effort: 158N of Theoretical force at 6bar, retracting (loading the charge)



Figure 33:DSNU -20-250-P-A [8]

2.1.2. Electromagnet lock

This electrical component is responsible for holding and releasing crimp applicators (ferrous material) on command. The main role is to guarantee the safety of tools while transportation. The major considered criteria to choose electromagnet are: holding force, component

dimensions and electrical requirements: the breaking power of this contact can't exceed 24V DC and 2A.



Figure 34:EF300CTC electromagnet lock

2.1.3. Motors specifications

Asynchronous motors are commonly used in industry due to their higher flexibility in many features including speed control using variable speed drivers. Three-Phase induction motors are widely used in industrial applications including compressors, pumps, conveyor systems and lifting gear.

AC motors have a much larger installed base compared to DC motors, some of the key advantages are:

- Low maintenance frequency,
- Low power demand on start,

- Controlled acceleration,
- Adjustable operational speed,
- Controlled starting current,
- Adjustable torque limit,
- Reduced power line disturbance.

Three phase motors asynchronous are consequently chosen for Z and X axis. Positioning characteristics are given in the **table 12** below:

Table 11: Positioning information

	Symbol	Unit	X	Z
Mass	M	Kg	141	45,5
Speed Movement	V	$mm.s^{-1}$	800	500
Distance Travel	D	Mm	8000	2350
Accelerating Time	T	S	3	1
Number of pitch /round	Z		32	20
Number of phases	n1		2	2
Diameter	D	M	0,1	0,05
Pitch	Pas	M	0,01	0,01

Using data in table 12 leads to listing the expressions and numeric values of key characteristics in table 13 to choose the convenient motors for X and Z axis:

Table 12: Motors dimensioning

	Expression	Symbol	Unit	X	Z
Force	m.g	F	N	1383,21	446,36
belt / rack efficiency		P		0,90	0,90
Speed motor	$\frac{V}{D.\pi}$	n	Rps	2,60	3,20
Speed motor	60.n	N	Rpm	**158,00**	**191,00**
Acceleration time		a	m/s/s	0,26	0,50
Trained Force	F.a	Ft	N	36,66	22,75
Torque	0,5.D.Ft	M	N.m	1,83	0,57
Friction coefficient		f		0,5	0,3
Friction torque	0,5.D.F.f	Mf		34,58	3,35
Total torque	M+Mf	Mt	N.m	36,41	3,92
Used Power	Mt.n.2Π	P	N.m	**594,86**	**78,74**

Motor datasheet, presented in the appendix 6 [9], details their constructor specifications for Z

axis.

2.2. Pre-actuators specifications

2.2.1. Pneumatic distributor specifications

The pneumatic chosen cylinders are double-acting therefore the distributor should have two orifices on which it is necessary to alternate the pressure and exhaust conditions. Thus, the adapted pre-actuator is a five-port 5/2 valves and two positions distributor.

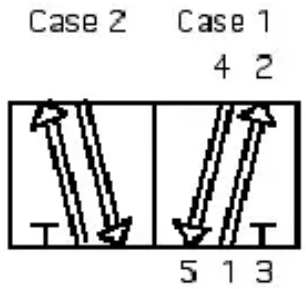


Figure 35: 5/2 distributor

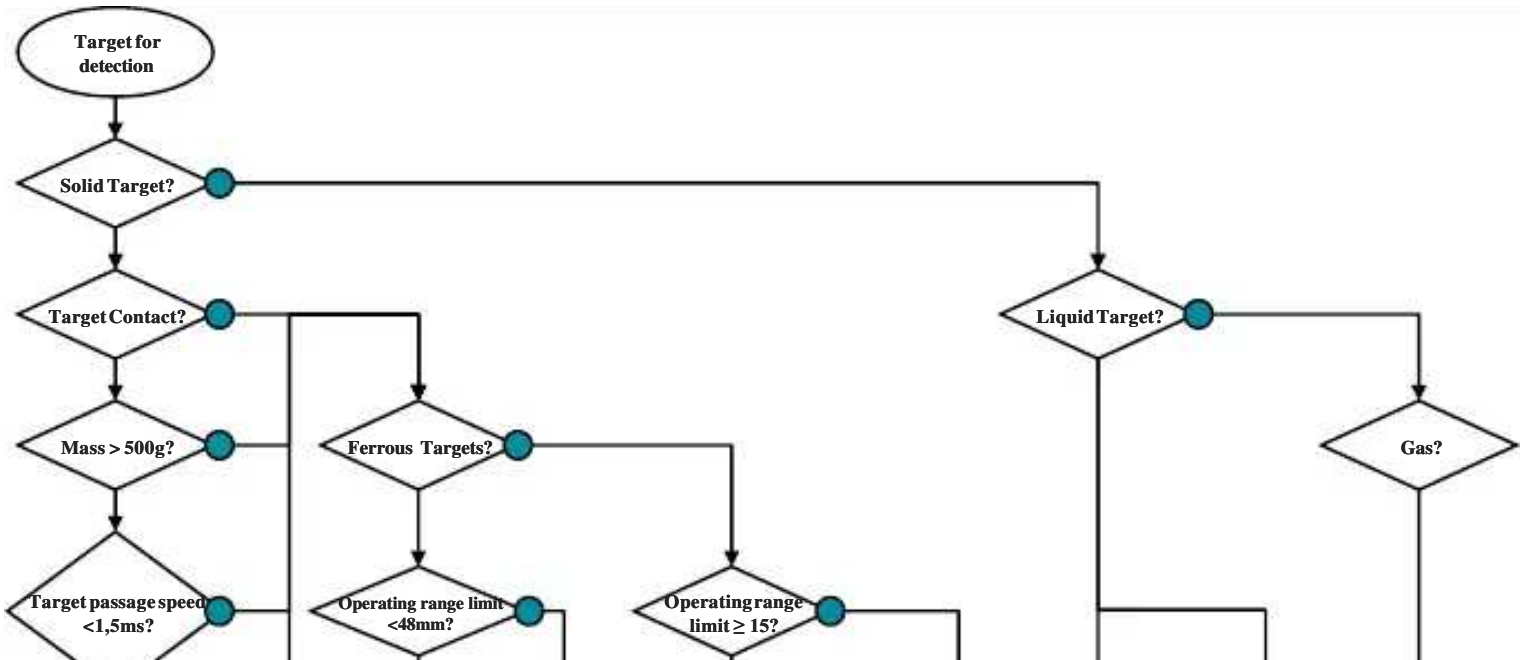
In absence of control signal, the distributor should memorize the state. Therefore, a bistable electrically commanded distributor is recommended.

2.2.2. Variable speed driver specifications

SINAMICS G120 variable speed driver is selected to control the speed asynchronous motors for Z-Axis and X-Axis. It communicates with SIMATIC S7300 via PROFIBUS DP.

2.3. Switching Sensors

While choosing sensors, many factors should be taken into consideration: the constitution of the target (material), existing contact with the target, the mass of the target, the frequency and speed of its passage and the environment condition (vibrations, temperature, pressure…). The methodology is detailed in the diagram in Figure 36.



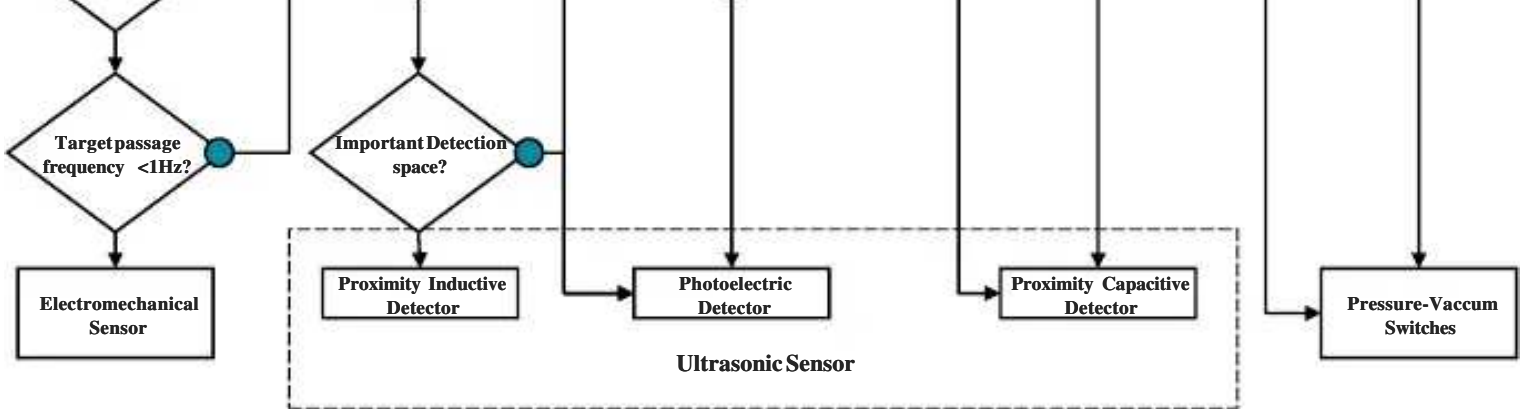


Figure 36: Choosing sensors Methodology

2.3.1. Proximity Inductive sensor

The four storage and retrieval zones of the racks reachable by technicians are equipped with inductive sensors to detect the presence of crimp applicators (metallic tools) in the box. Supported current, Voltage and other technical data are indicated in the appendix 7.

2.3.2. Pneumatic cylinder position switches ILS

The activation of the electromagnet lock depends on the position IN and OUT of the pneumatic cylinders. So, cylinder position switches are the suitable sensors to be directly mounted on the actuator.

2.3.3. Optical distance measurement sensor

It is a high precision distance sensor with large operating range and accuracy compared to the encoder. So, it is selected to measure the stacker crane moving according to X axis and Z axis. The emitter is positioned at the end of travel on a fixed part while the reflector on board stacker cranes is mobile and projects its laser beam on a reflector to measure the distance. The datasheet in appendix 8 specifies the voltage, current and the ODS laser sensor operating range.

3. Process and Control specifications

3.1. PLC characteristics

The PLC is chosen in the optical to standardize the equipment used within the DRX. group, namely the SIEMENS brand. A wide range of Siemens PLC's is commercially available: SIMATIC S7-200, SIMAIC S7-300, SIMATIC S7-400, SIMATIC S7-1200, SIMATIC S7-1500. However, there are many distincting points:

- Basic Instructions
- Extended Instructions
- Technology

- Communication

A comparative table was given by Siemens to facilitate the distinction and the choice for customers. To guarantee the detection of the vacant place for a returning tool and the access to an already stored tool, the PLC needs the specific table instructions as presented in the table below:

Table 13: S7 PLC's Comparison [10]

S7-300	S7-400	S7-1200	S7-1500	Table functions	
✓	✓			Add value to table	ATT
✓	✓			Output first value of the table	FIFO
✓	✓			Find value in table	TBL_FIND
✓	✓			Output last value of the table	LIFO
✓	✓			Execute table instruction	TBL_FIND
✓	✓			Copy value from table	TBL_WRD
✓	✓			Link value logically with table element and save	WRD_TBL
✓	✓			Calculate standard deviation	DEV
✓	✓			Correlated data tables	CDT
✓	✓			Link tables	TBL_TBL
✓	✓			Collect/distribute table data	PACK

S7-300 and S7-400 offers both the advantage of table manipulation.

S7-400 are made for complex and large machinery system. It is a high-level PLC whereas it is oversized for the automated stacker crane.

Therefore, the selected PLC is S7-300. It offers a scaled range of 24 CPUs; from standard processors to the first mainframe with integrated Ethernet / PROFI net interface, security processors, compact processors with integrated technology and peripheral functions, and technology processors for command management.

The S7-300 also offers a very large range of digital and analog I / O modules for virtually all signals with the possibility of interrupt processing and diagnoses.

- **Recommended Input/Output**

Table 14: INPUT/OUTPUT classification

Type of INPUT /OUTPUT	Voltage/Current	Numbers of INPUT/OUTPUT	Number IO recommended
Relay INPUT	24Vcc	16	20
Analog INPUT	4..20mA	2	3

Relay OUTPUT	24Vcc	17	22
Analog OUTPUT	+ or – 10V	4	5

$Number\ recommended = Number\ of IO + 25\%$

- **Power Supply is referenced:** 6ES7 307-1KA02-0AA0

External Supply: 120/230 V ca: 24 V cc/10 A

- **CPU : 6ES7 315-2AF00-0AB0**

It is characterized by:

48 Ko Memory; 0,3ms/kinst.; MPI+DP ports; multi-row configuration up to 32 modules

3.2. HMI characteristics

Human machine interfaces are used for visualization as well as for machine control. They allow to collect the I/O data of a PLC and to present them in a way that makes them comprehensible and exploitable by the user.

In Siemens environment, HMIs are programmed with **WinCC Flexible 2008** or with **TIA Portal**. However, there is another way to supervise installations or machines through the design

of a specific application in computer language. Create a specific supervision application eliminates the purchase of software sales supervisors like **WinCC, Vijeo Designer** etc.

Moreover, designing a specific supervision application can save money, because commercial software usually cost quite expensive and their prices can reach several thousand euros. Nowadays, it is quite common to find manufacturers who combine an OPC server and Visual Studio / WinDev to design applications via C # or VB languages.

Therefore, visual studio IDE is chosen to develop Stacker cranes human machine interface.

- **Microsoft Visual Studio**

It is an integrated development environment (IDE) from Microsoft (Figure 37). It can be used to develop console and graphical user interface applications along with Windows Forms applications, web sites… [11]

- **C#**

It is an object-oriented programming language and part of the .NET family from Microsoft (Figure 38). C# is very similar to C++ and Java. C# is developed by Microsoft and works only

on the Windows platform [11]



Figure 37: Visual C#

4. Electrical installation

The electrical installation is structured as below:

- Electrical power connection which includes the actuators (motors, electromagnetic locks), the contactors. It is illustrating in the appendix 9
- Electrical command connection which includes the inputs and outputs electrical connection

Conclusion

In this chapter, I explained at first the mechanical chosen solution for stacker cranes then I presented the instrumentation methodology to guarantee the security and the precision of the system. In the next chapter, I will describe the functioning modes and the control through the HMI.

Chapter 4: Stacker Crane Automation and HMI

Automation and HMI Development

Introduction

The fourth chapter is aimed to presenting the last part of stacker crane feasibility study. It is carried out after the mechanical and electrical architecture definition to detail the functioning mode of the proposed system and the control through the human machine interface.

1. Automation

It is necessary to define different operating and stopping modes of the installation through the general description and hardware and network setting.

1.1. General description

Three different functioning modes are set to manage stacker cranes movement. The following "GRAFCETs" illustrate process management of:

- Chosen Operating mode,
- Defects
- Automatic Operating Mode.

The Figure 38 illustrates the management of different modes (Manual, Automatic or reference setting). Manual mode consists in manually selecting the storage location or consulting the rack

image to retrieve the tool.

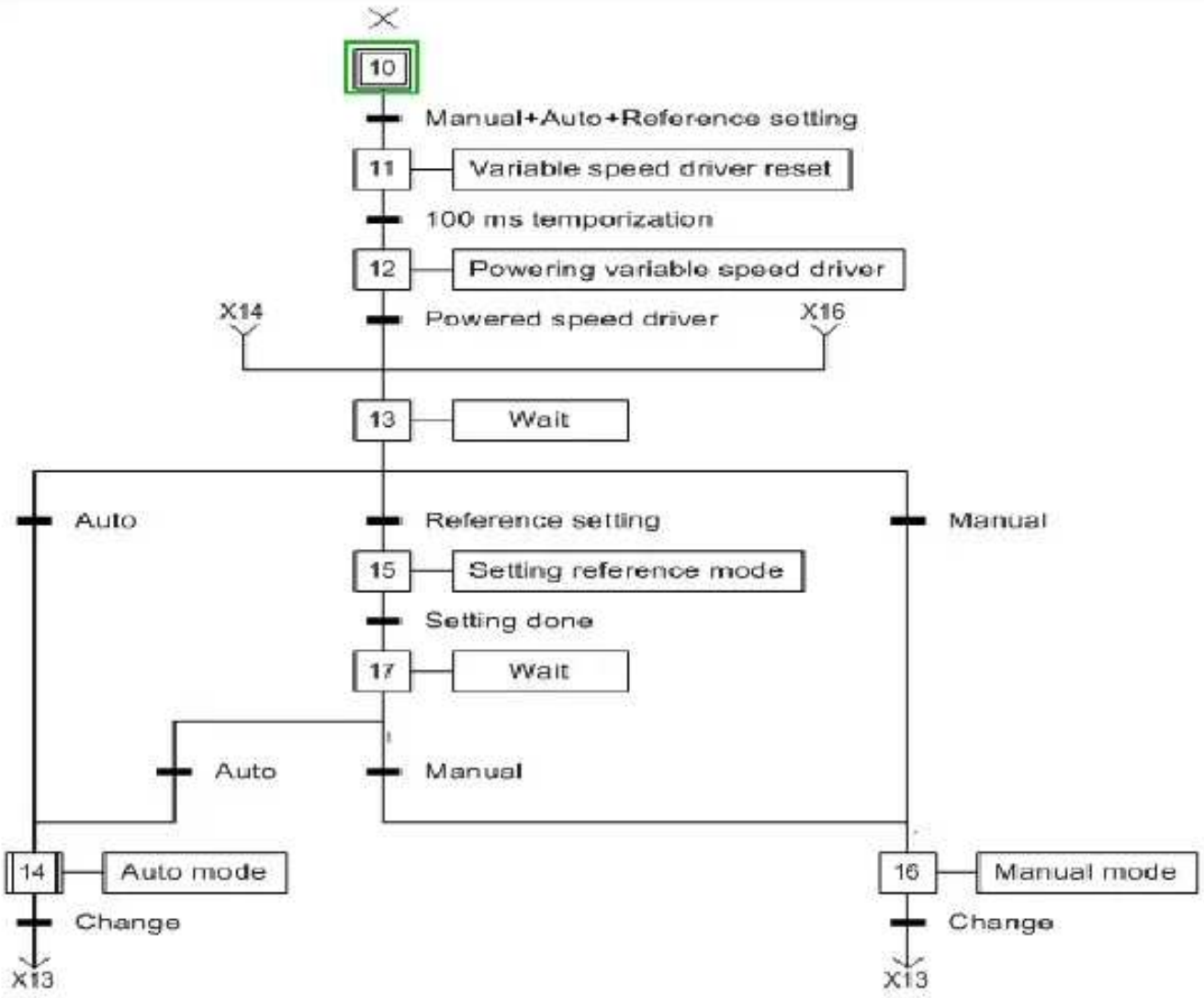


Figure 38: Management of different Modes

Figure 39 presents the defect management. If a technical problem occurs, such as a non-rotation of motors or , system Reset starts until eliminating the source of the problem.

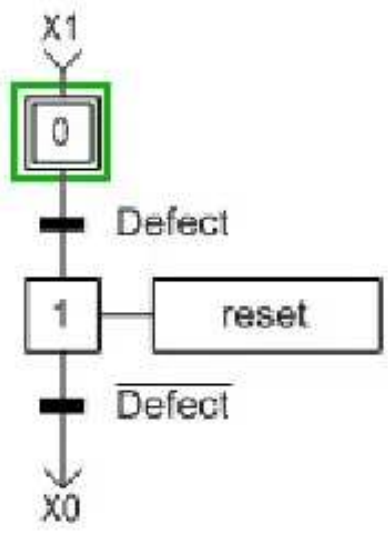


Figure 39: Security "GRAFCET"

The Figure 40 illustrates the validation of the storage and retrieval of crimp applicators. If they do not exist in the box or there are no available vacant boxes, or the stacker crane is occupied, an error message is displayed, and the technician should key in the reference again after only 4 seconds.

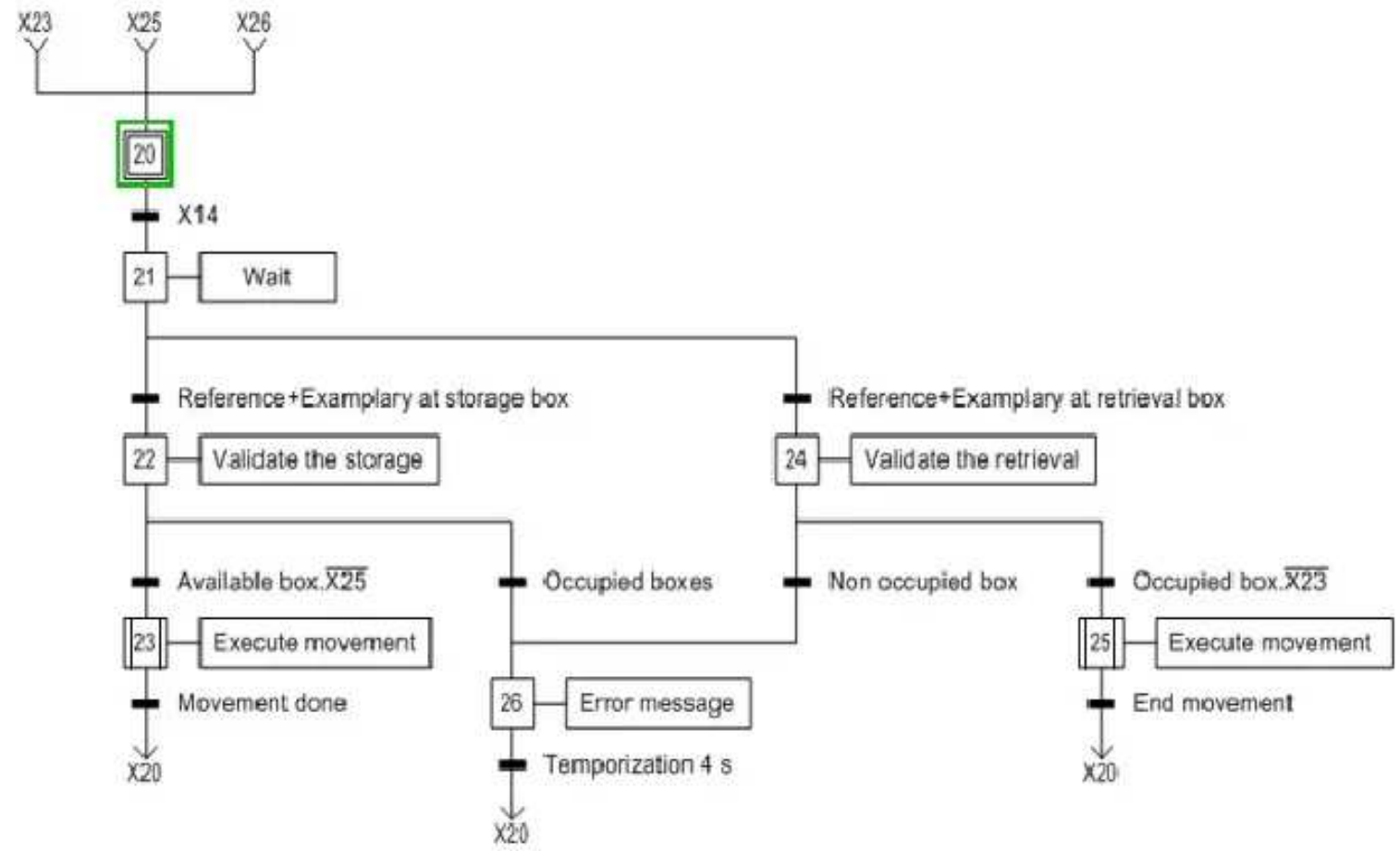


Figure 40: Management of Storage and retrieval GRAFCET

The figure 41 illustrates the execution of the movement all over the three axes, the stacker cranes needs to go back to the starting position to start a new cycle of retrieval or storage.

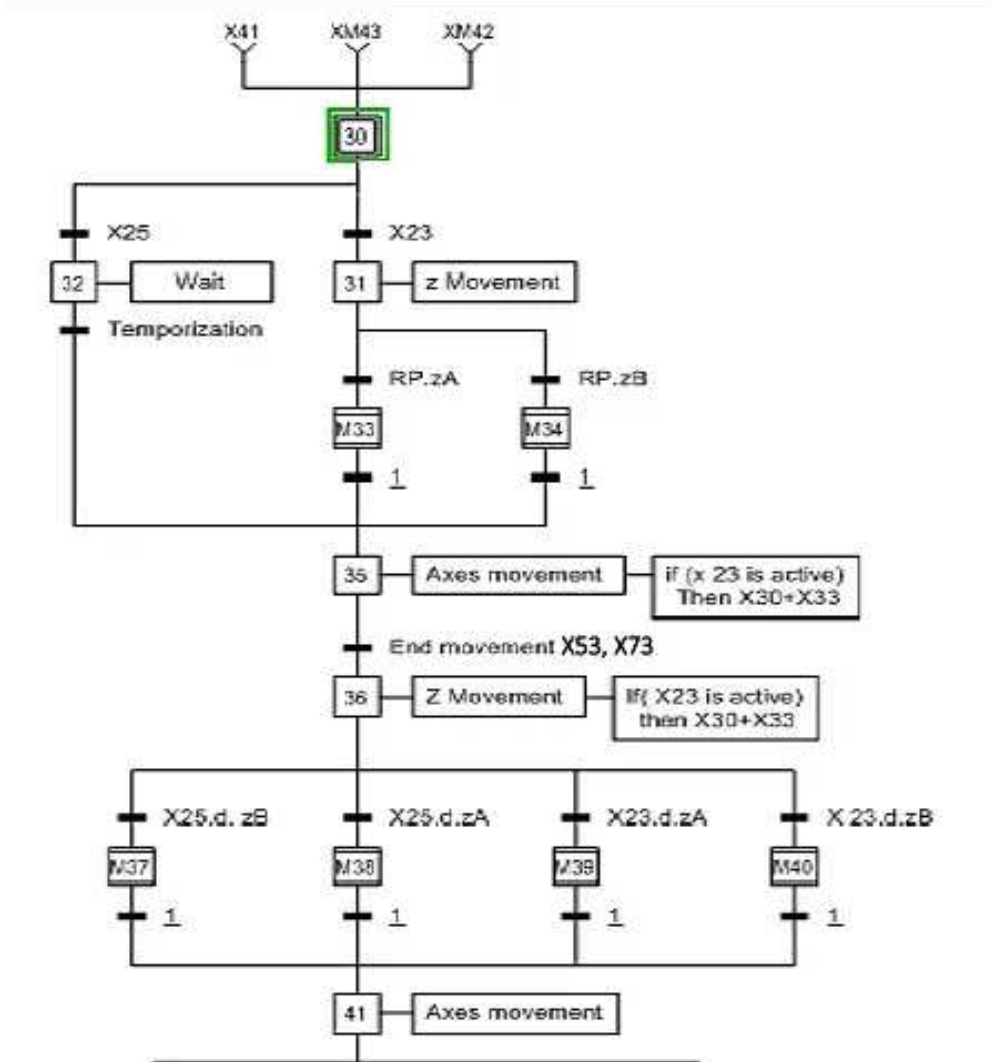




Figure 41: Operating tools retrieval and storage

The abbreviations used in the Figure 41 are explained in the following list:

RP	: Reached position	SD	: Storage done
ZA	: A zone	RD	: Retrieval done
ZB	: B zone		

X53, X63, X73 and X83 are leading to the activations of the variable speed driver for Z axis and Z axis.

Macro steps are developed in the Figure 42 to describe the Y axis operating mode while commanding the electromagnet lock and the pneumatic cylinder.

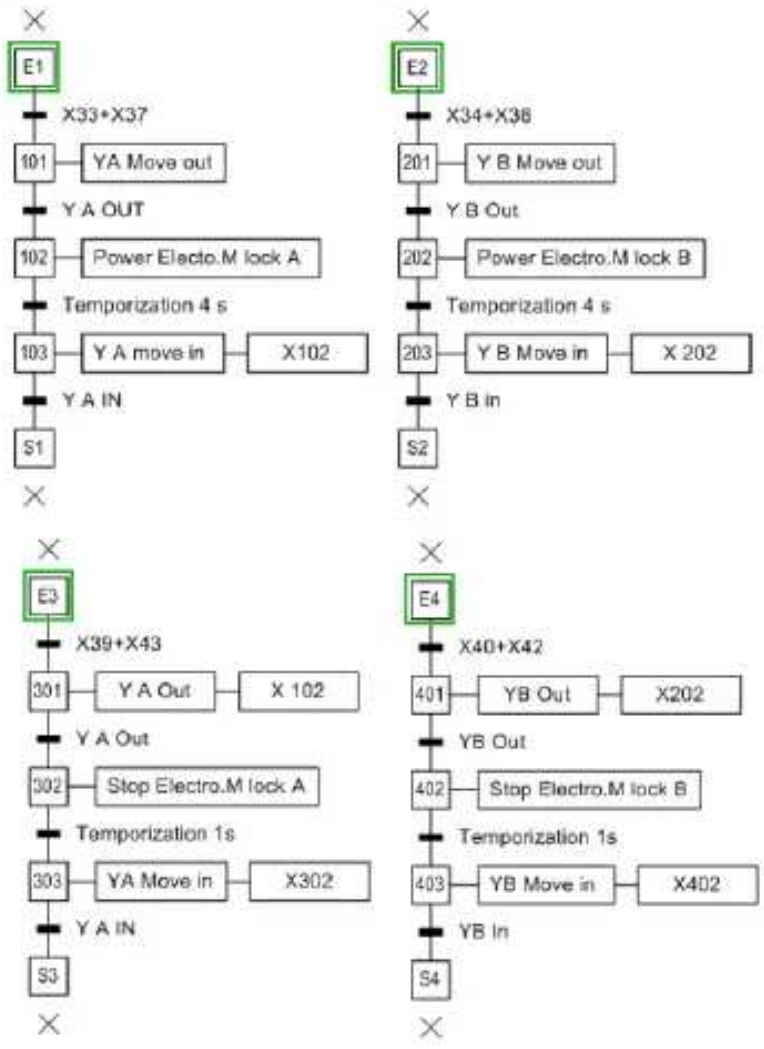


Figure 42: Developped macro-steps

1.2. Hardware Configuration

The necessary number of INPUT/ OUTPUT Deported Modules are detailed in the list below:

Table 15: Recommended Input and Output modules

Number of Module	Type	I/O	Number	characteristics
1	D	I	16	24Vcc1
1	A	O	8	12Bit
1	A	I	4	4..20mA
1	D	O	16	24Vcc/0,5A
1	D	I/O	8/8	24Vcc/0,5A

The following Figure 43 details the number of INPUT and OUTPUT modules and indicates the corresponding addresses for PLC programming phase to facilitate the development phase.

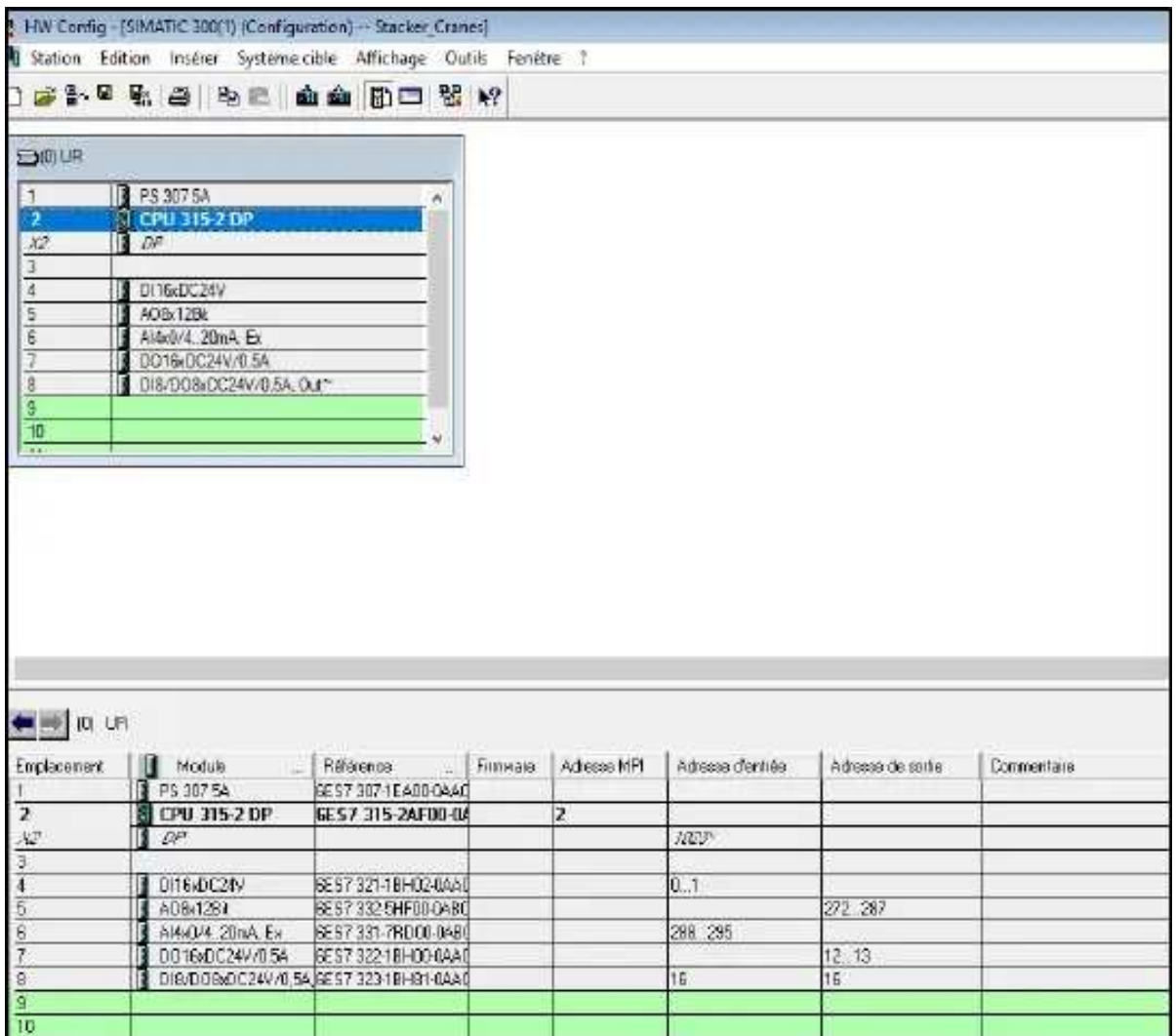




Figure 43: Hardware configuration

1.3. Network configuration

SIMATIC S7-300 is controlling the variable speed driver via PROFIBUS DP.

The controller is to have read and write access to process data to change update values. The faults and alarms are detected and displayed on the operator HMI as shown in Figure 44.

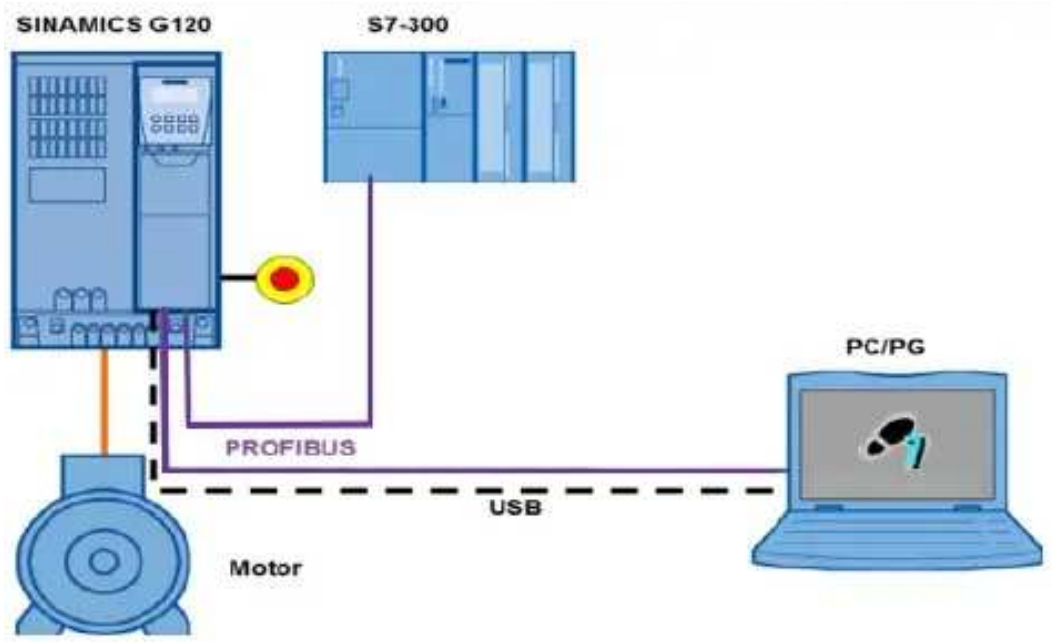


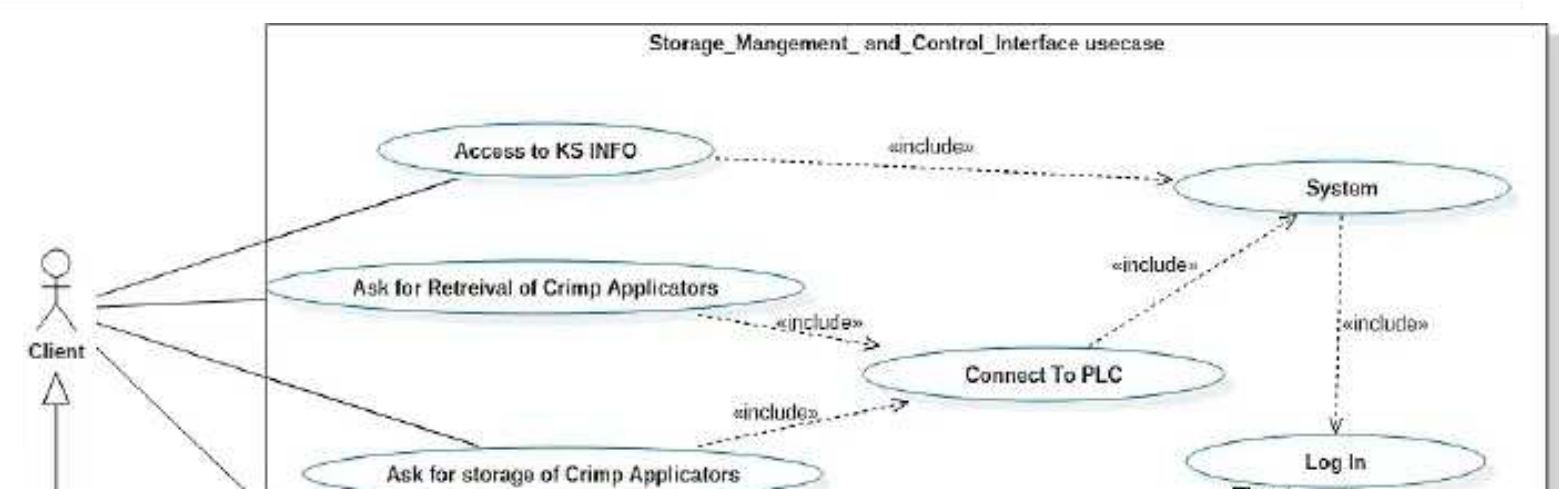
Figure 44: Network configuration

2. HMI Conception

The Human Machine Interface guarantees the management of defects and safeties. It allows technicians to manage the retrieval and storage

2.1. Stacker Crane Use Case Diagram

After analyzing the requirement specifications of the warehouse and to provide better flexibility in management of crimp applicators, the use case diagram illustrates the global functionalities of the HMI, presented in the following figure 45:



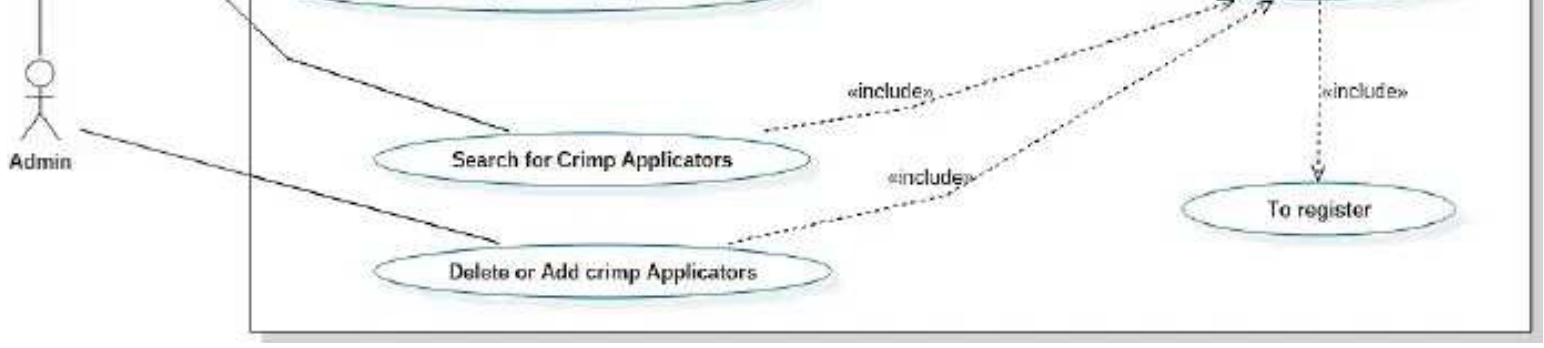


Figure 45: Use Case Diagram

Uses case description:

- Use Case: Ask for Crimp Applicators Retrieval

Title	Retrieval
Actor	Client
Pre-condition	PLC_Connetion
Principal scenario	• The user launches the application • The user logs in or signs in if he is not registered • The user chooses “Control_System” option • The user connects to the PLC • The user enters the reference of tool and its exemplary which should be already existing in the database • The user validates the retrieval by clicking on the button

- Use Case: Ask for Crimp Applicators Storage

Title	Storage
Actor	Client
Pre-condition	PLC_Connetion
Principal scenario	• The user launches the application • The user logs in or signs in if he is not registered • The user chooses “Control_System” option • The user connects to the PLC • The user enters the reference of tool and its exemplary which should be already existing in the database • The user validates of retrieval by clicking on the button

- Uses Case: Connect to PLC

Title	PLC_Connection
Actor	Client (Technicians)
Pre-condition	System_Control
Principal	• The user launches the application

scenario	• The user logs in or signs in of he is not registered • The user chooses “Control_System” option • The user clicks on the button PLC_Connection to connect the application to the PLC

- Use Case: Access to KS-INFO

Title	KS_INFO
Actor	Client
Pre-condition	System_Control
Principal scenario	• The user launches the application • The user logs in or signs in of he is not registered • The user chooses “Control_System” option • The user clicks on the button PLC_Connection to connect the application to the PLC

- Use Case: Search for crimp applicator

Title	Search
Actor	Client
Pre-condition	Log In / Sign In
Principal scenario	• The user launches the application • The user logs in or signs in of he is not registered • The user chooses “Search” option • The user chooses the criteria of sorting and enters the value to search

- Use Case: Delete and Add Tools

Title	Delete /Add
Actor	Admin (Warehouse Responsible)
Pre-condition	Log In
Principal scenario	• The user launches the application • The user logs in or signs in if he is not registered • The user chooses to delete or add a tool by reference and